

# A Hardware-Efficient Hybrid Beamforming Architecture for Near-Field Beam Training in 6G Wideband XL-MIMO Systems

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**ABSTRACT** Near-field wideband beam training is a key bottleneck in millimeter-wave (mmWave) and terahertz (THz) XL-MIMO systems, because channel acquisition depends jointly on angle and range, while the wide bandwidth introduces frequency-dependent beam split. Notably, such frequency-dependent beam splitting can also be exploited to enable fast near-field beam training. Existing distance-dependent beam-split (DDBS) training can exploit subcarriers to search multiple angles and distances simultaneously in a single time slot; however, it typically relies on employing true-time-delay (TTD) hardware with every antenna element, which is costly and power-consuming. This paper proposes a hardware-efficient array-of-sub-arrays (AoSA)-based true-time-delay/phase-shifter (TTD-PS) architecture, where one TTD is shared within each antenna sub-array (one TTD per sub-array). We characterize how delay sharing reshapes DDBS behavior by restricting the strong-gain region to a limited angular range/sector. We adopt practical 3-dB coverage rules that determine the required number of pilots to cover the entire angular range and distance domain, yielding an analytical expression for the training overhead. Building on these results, we develop a sector-shifted and distance-interleaved training procedure with least-squares (LS)-based focusing refinement. We also propose an off-grid match-filter method to improve robustness under noise and grid mismatch. It is analytically shown that under the AoSA-based TTD-PS architecture, a few pilots (e.g., three pilots, as shown in prior per-antenna TTD work for per-antenna TTD hardware) are not sufficient to cover the whole angle-distance space for wideband XL-MIMO. Simulations show that with a single TTD shared by two antennas, the proposed design achieves a favorable tradeoff, halving the number of TTD elements while approaching the performance of the ungrouped architecture with about 12 pilots.

**INDEX TERMS** Extremely large-scale MIMO (XL-MIMO), near-field communications, wideband systems, hybrid beamforming, beam training, grouped true-time delay (TTD), array-of-sub-arrays (AoSA), distance-dependent beam split.

## I. INTRODUCTION

THE push for performance far beyond what current 5G systems can deliver is accelerating the transition toward sixth-generation (6G) wireless networks, which are expected to support post-2030 requirements and enable new applications through Tb/s-level throughput, ultra-low latency, and high reliability. These ambitious targets motivate much wider bandwidths and much larger antenna apertures than in conventional massive MIMO, making extremely

large-scale MIMO (XL-MIMO) a key enabling technology for future networks [1], [2], [3], [4], [5]. Unlike conventional arrays where far-field assumptions are typically valid, XL-MIMO fundamentally changes the electromagnetic behavior of wireless propagation. As the physical aperture grows, the Rayleigh distance increases significantly, and many practical users fall within the radiative near-field region. In this regime, the plane-wave approximation no longer holds and the channel must be modeled using spherical

wavefronts; consequently, the array response depends jointly on angle and range, which calls for new near-field signal processing techniques [3], [6].

A central implication is that beam training and channel state information (CSI) acquisition become substantially more complex. Rather than angle-only alignment, the transmitter must localize the user in an angle-distance space, and wideband operation introduces frequency-dependent beam split across subcarriers. While true-time-delay (TTD) aided analog beamforming can control (and even exploit) this frequency dependence, per-antenna TTD realizations are costly and difficult to scale. This motivates near-field wideband beam training designs that explicitly incorporate scalable hardware constraints, particularly sub-array-shared TTDs in array-of-sub-arrays (AoSA) architectures [7], [8], [9], [10], [11], [12], [13].

### A. RELATED WORKS

In far-field wideband systems, frequency selectivity in analog beamforming leads to beam squint/beam split across subcarriers [7], [8]. To address this issue, true-time-delay (TTD) elements and frequency-dependent beamforming have been investigated, including delay-phase/joint delay-phase style architectures for wideband beam shaping [7], and more general joint time-phase control frameworks such as joint phase-time arrays (JPTA) [10]. In addition, hierarchical designs (e.g., spatial-chirp codebook based training) can reduce training overhead in large arrays [9]. Beyond initial alignment, wideband beam tracking under beam split has also been studied using delay-phase precoding structures, where beam-zooming and subcarrier-dependent beams are used to efficiently track the dominant direction in wideband THz massive MIMO with reduced tracking overhead [14].

In XL-MIMO, many users operate in the radiative near-field region, where the channel depends jointly on angle and range rather than angle alone [6], [15]. In narrowband systems, this hybrid-field propagation makes beam training a joint angle-range estimation problem. Polar-domain codebooks that sample the full angle-distance plane can enable accurate alignment, but exhaustive search over the resulting 2D grid incurs prohibitive training overhead due to the curse of dimensionality [15]. To reduce pilots, hierarchical and refinement-based near-field beam training methods have been developed. Chen et al. proposed a two-stage hybrid-field beam training/tracking framework with phase-shift refinement [16] and further introduced a triple-refined scheme to improve accuracy under limited overhead [17]. Under the narrowband condition, Lu et al. presented a hierarchical far-field-to-near-field training method using multi-resolution near-field codebooks under hybrid constraints [18]. Related approaches include spatial-chirp based hierarchical search [9], two-stage schemes combining coarse far-field angle estimation with near-field refinement [19], and coarse-angle then fine-distance estimation strategies [20]. Mobility-oriented beam training/tracking has also been studied [16], [21].

Wideband near-field training is further complicated by beam split across subcarriers; however, this frequency dependence can also be exploited for faster localization. The near-field rainbow concept [22], [25] uses tuned delays so that different subcarriers focus on different angles within a distance ring (i.e., frequency-division angular search). Nevertheless, it does not resolve range without a sweeping procedure and typically assumes idealized per-antenna true-time-delay (TTD) hardware, thereby requiring distance sweeping. More recent works explicitly leverage near-field wideband beam split trajectories for joint angle-range inference, e.g., deep learning aided localization with subcarrier grouping for refinement [24]. The most relevant recent advance is distance-dependent beam split (DDBS) training, which jointly designs delay and phase-shift parameters to generate multi-strip patterns in the angle-range domain and probes candidate angles and distances with only a few pilots [23].

Although DDBS-type approaches provide very low training overhead, they are typically developed under per-antenna TTD assumptions (one tunable delay per antenna), which is difficult to scale to XL-MIMO due to cost, power consumption, routing, and calibration complexity [23]. In contrast, AoSA architectures are widely considered for large apertures, where delay elements can be shared at the sub-array level [11], [12], [13]. However, existing AoSA designs are not tailored to realize DDBS-style multi-strip angle-range probing under sub-array-level delay sharing. Related throughput-oriented efforts also share/group TTD resources across antenna groups. For example, beamfocusing optimization for near-field wideband multi-user communications also considers sub-connected TTD-PS hybrid architectures to reduce hardware constraints and improve energy efficiency [26]; JPTA enables antenna clusters to share a single TTD while supporting frequency-dependent beamforming [10]; dynamic-sub-array designs with fixed TTD mitigate beam squint using low-cost delays and switching [11]; delay-phase-precoding based beam tracking uses a limited number of time-delayers per radio-frequency (RF) chain shared across groups while retaining subcarrier-dependent beam control [14]; and quantization-aware designs show TTD-assisted hybrid beamforming remains effective even with coarse hardware [27]. These studies also indicate that TTD should be viewed as a practically motivated hardware enabler for wideband/mmWave/THz beam control rather than a purely idealized assumption. At the same time, these studies consistently suggest that fully tunable per-antenna TTD implementations are difficult to scale to XL-MIMO because of the associated hardware cost, power consumption, routing complexity, and calibration burden [28]. This is precisely why grouped or sub-array-shared TTD architectures are of practical interest: they retain the key frequency-dependent beam-control capability needed for wideband operation while significantly reducing the number of expensive delay elements. However, these works mainly focus on beam focusing, beamforming, and beam tracking in wideband communication systems, where the objective is

to mitigate beam split using TTDs to improve the received signal-to-noise ratio (SNR). In particular, several of these studies proposed hardware-efficient architectures to reduce the cost and complexity of implementing wideband focusing and tracking. However, they do not address the design of near-field wideband beam training, which requires efficiently probing the joint angle–distance space with limited pilots. Although this problem has been recently addressed in [23], their architecture is hardware-inefficient, costly, and power-hungry, due to the reliance on a TTD element for every antenna element. Our work addresses this gap by proposing a hardware-efficient hybrid beamforming architecture and near-field beam training design in 6G wideband XL-MIMO systems.

## B. OUR CONTRIBUTIONS

Motivated by the gap between the training efficiency of distance-dependent beam split (DDBS) and the scalability limits of practical XL-MIMO hardware, we propose an AoSA-based TTD–PS architecture, where a single TTD element is shared by multiple antennas in each sub-array. Our aim is to enable practical angle–range probing, even in the presence of a reduced number of TTD elements. The main contributions are summarized as follows:

- **DDBS Beam Training With Sub-array-Shared TTDs:** We extend DDBS to an AoSA TTD–PS implementation by enforcing a grouped-delay constraint, where a single TTD is shared within each sub-array comprising multiple antennas. This significantly reduces the delay-element count while preserving frequency-dependent probing and enabling efficient near-field angle–distance training via *grid-based (codebook-based) estimation*.
- **Grouping-aware near-field wideband beam model:** We develop a near-field wideband beam model for the adopted sub-array architecture and characterize the grouping-induced angular–range periodicities and their impact on beam focusing across different subcarriers. These properties provide design guidelines for effective beam training with a reduced number of TTDs.
- **Grouped Training With Design Rules:** We show that under sub-array-shared TTDs, the strong-gain region is confined to a limited angular sector. As a result, few pilots, which were shown sufficient under per-antenna TTD hardware, are not adequate to guarantee full angle–distance coverage over the considered search region (i.e., the target angular range and distance range). Building on the grouping-induced periodic focusing/sectoring characterization, we propose a sector-shifted and distance-interleaved training method for near-field wideband XL-MIMO. We develop a 3-dB coverage criterion, which provides practical parameter and pilot-selection rules, and quantifies the required training overhead under grid-based (codebook-based) estimation.
- **Robust Refinement and Performance Validation:** To mitigate noise and *codebook mismatch due to finite*

*resolution*, we also incorporate a match-filter-based refinement for the AoSA TTD–PS architecture. Simulations show that near-optimal spectral efficiency is achieved with a practical, analytically determined pilot budget, while substantially reducing the number of delay elements across a wide range of SNRs and user distances.

A comparison of existing near-field training methods is provided in Table 1.

## C. ORGANIZATION AND NOTATION

*Organization:* The rest of this paper is organized as follows. Section II presents the near-field wideband XL-MIMO system model and the proposed AoSA TTD–PS architecture with grouped (shared) TTD elements and formulates the beam training problem. Section III analyzes the array gain behavior under sub-array-shared TTDs and characterizes the resulting wideband focusing/beam-split effects, including the key angle–range periodicity properties. Section IV develops the sectoring phenomenon and the periodic focusing laws under the grouped TTD–PS structure, which serve as the basis for coverage-aware beam and parameter design. Section V proposes the sub-array distance-dependent beam-split (DDBS) beam training framework, including the multi-strip training construction, sector-shift design, distance interleaving, and the associated pilot/parameter selection rules. Section VI presents a robustness refinement via match-filter enhancement to mitigate noise sensitivity and grid mismatch. Section VII provides simulation results and comparisons with baseline methods. Section VIII concludes the paper.

*Notations:* Scalars are denoted by italic letters, vectors by bold lowercase letters, and matrices by bold uppercase letters. The operators  $(\cdot)^T$ ,  $(\cdot)^*$ , and  $(\cdot)^H$  denote transpose, complex conjugate, and conjugate transpose, respectively.  $\|\cdot\|_2$  denotes the  $\ell_2$ -norm,  $\|\cdot\|_F$  denotes the Frobenius norm, and  $|\cdot|$  denotes the absolute value (modulus).  $\text{diag}(\cdot)$  forms a diagonal matrix, and  $\odot$  denotes the Hadamard product.  $[\mathbf{x}]_i$  denotes the  $i$ th entry of vector  $\mathbf{x}$ , and  $[\mathbf{A}]_{i,j}$  denotes the  $(i,j)$ th entry of matrix  $\mathbf{A}$ .  $\mathbb{E}[\cdot]$  denotes expectation, and  $\mathcal{CN}(\mu, \Sigma)$  denotes a circularly symmetric complex Gaussian distribution with mean  $\mu$  and covariance  $\Sigma$ .  $\lceil \cdot \rceil$ ,  $\lfloor \cdot \rfloor$ , and  $\text{round}(\cdot)$  denote ceiling, floor, and rounding to the nearest integer, respectively.  $\mathbf{I}_N$  denotes the  $N \times N$  identity matrix.

## II. SYSTEM MODEL AND PROBLEM FORMULATION

This section introduces the near-field wideband XL-MIMO channel model and its polar-domain parameterization. Then, we describe the AoSA TTD–PS architecture with sub-array TTD elements and formulate the beam training observation model and objective.

### A. SIGNAL AND CHANNEL MODELS

We study a downlink wideband near-field XL-MIMO system operating at mmWave and THz frequencies where the base station (BS) employs a uniform linear array (ULA) with an odd number of antennas, denoted by  $N_t$  for notational

simplicity, and serves a single-antenna user equipment (UE) using orthogonal frequency-division multiplexing (OFDM) with  $M$  subcarriers. The system bandwidth and carrier frequency are denoted by  $B$  and  $f_c$ , respectively. The speed of light is  $c$ , the central wavelength is  $\lambda_c = c/f_c$ , and the antenna spacing is  $d = \lambda_c/2$ . Let  $f_L$  and  $f_H$  denote the lower and upper subcarrier frequencies, defined as  $f_L = f_c - \frac{B}{2}$  and  $f_H = f_c + \frac{B}{2}$ . With uniform spacing  $\frac{B}{M}$ , the  $m$ -th subcarrier frequency becomes  $f_m = f_L + \frac{B}{M}(m-1)$ .

For notational convenience, symmetric indexing is used for the antenna elements, i.e.,  $\hat{n} \in \{-\frac{N_t-1}{2}, -\frac{N_t-1}{2}+1, \dots, \frac{N_t-1}{2}\}$ . Equivalently, the physical antenna elements can be indexed by  $n \in \{0, 1, \dots, N_t-1\}$ , and the centered index is obtained by shifting the origin to the array center, i.e.,  $\hat{n} = n - \frac{N_t-1}{2}$ . Hereafter,  $\hat{n}$  denotes the centered antenna index and will be used to simplify the near-field phase expressions.

We consider downlink training transmission. On subcarrier  $m$ , the BS applies the beamforming vector  $\mathbf{v}_m$  and transmits a normalized pilot symbol  $\psi_m = 1$ . The received sample at the UE is modeled as

$$\chi_m = \sqrt{P_t} \mathbf{g}_m^H \mathbf{v}_m \psi_m + v_m, \quad (1)$$

with transmit power  $P_t > 0$ , channel vector  $\mathbf{g}_m \in \mathbb{C}^{N_t \times 1}$ , and unit-norm beamforming vector  $\mathbf{v}_m \in \mathbb{C}^{N_t \times 1}$ . The noise term  $v_m$  follows  $v_m \sim \mathcal{CN}(0, \sigma^2)$ .

A Saleh-Valenzuela wideband channel model [29] is adopted, which is well suited for mmWave and THz frequencies where propagation is characterized by a sparse set of clusters. The downlink channel at subcarrier  $m$  is generally expressed as

$$\mathbf{g}_m = \sqrt{\frac{N_t}{P}} \sum_{\ell=1}^P \zeta_{m,\ell} e^{-j2\pi f_m \tau_\ell} \mathbf{u}_m(\theta_\ell, r_\ell), \quad (2)$$

where  $P$  is the number of paths,  $\zeta_{m,\ell}$  and  $\tau_\ell$  denote the complex path gain and delay, respectively, and  $r_\ell$  is the distance from the array center of the  $\ell$ -th path.

In near-field wideband XL-MIMO systems, the objective of initial beam training is to establish a reliable communication link by aligning the beam to the dominant propagation path, whether it is the line-of-sight (LoS) path or a strong specular reflection in non-LoS (NLoS) scenarios. Consequently, to facilitate the design and analysis of the proposed beam training scheme, the analytical derivations in Sections III–V focus on a dominant-path channel model, which is a special case of (2) with  $P = 1$ . In this case, the channel vector simplifies to

$$\mathbf{g}_m = \sqrt{N_t} \zeta_m e^{-jk_m r} \mathbf{u}_m(\theta, r), \quad (3)$$

where  $\zeta_m = \lambda_m/(4\pi r)$  represents the free-space path loss, and  $(\theta, r)$  denote the parameters of the dominant path. While the analytical results are derived under this dominant-path assumption, the performance of the proposed scheme under more realistic multipath channels with NLoS propagation is evaluated via simulations in Section VII.

To capture spherical-wave propagation in the near field, the distance from antenna element  $\hat{n}$  to the UE is

$r(\hat{n}) = \sqrt{r^2 + \hat{n}^2 d^2 - 2r\theta \hat{n}d}$ . Accordingly, the near-field array response  $\mathbf{u}_m(\theta, r)$  in (2) is

$$\mathbf{u}_m(\theta, r) = \frac{1}{\sqrt{N_t}} \left[ e^{-jk_m(r(-N)-r)}, \dots, e^{-jk_m(r(N)-r)} \right]^T, \quad (4)$$

where  $N = (N_t - 1)/2$ . Using a second-order (Fresnel) approximation of  $r(\hat{n}) - r$ , we have

$$r(\hat{n}) - r \approx -\hat{n}d\theta + \hat{n}^2 d^2 \alpha, \quad (5)$$

and

$$\alpha \triangleq \frac{1 - \theta^2}{2r}, \quad (6)$$

which is the wavefront-curvature (inverse-range) coefficient. Therefore,  $\mathbf{u}_m(\theta, r)$  can be approximated by the polar-domain steering vector

$$\mathbf{c}_m(\theta, \alpha) = \frac{1}{\sqrt{N_t}} \left[ e^{jk_m(\hat{n}d\theta - \hat{n}^2 d^2 \alpha)} \right]_{\hat{n}=-N}^N. \quad (7)$$

We thus represent the UE location by  $(\theta, \alpha)$ , which is equivalent to  $(\theta, r)$  since  $r = (1 - \theta^2)/(2\alpha)$ ; moreover,  $\alpha \rightarrow 0$  as  $r \rightarrow \infty$ , recovering the far-field steering vector.

## B. PROBLEM FORMULATION FOR NEAR-FIELD BEAM TRAINING

The objective of near-field beam training is to estimate the user location  $(\theta, \alpha)$  over a target angle–distance region using a sequence of pilot beamforming vectors  $\{\mathbf{v}_m^{(k)}\}_{k=1}^K$ .

Conventional near-field beam training treats this task as a grid-based matching problem over a predefined polar-domain codebook [15]. At the  $m$ -th subcarrier, the codebook is written as

$$\mathbf{C}_m = \left[ c_m(\theta_1, \alpha_1^1), \dots, c_m(\theta_{N_t}, \alpha_{N_t}^{S_{N_t}}) \right], \quad (8)$$

where  $S_n$  denotes the number of sampled distance grids associated with the sampled angle  $\theta_n$ .

Accordingly, the conventional exhaustive polar-codebook benchmark estimates the user location  $(\hat{\theta}, \hat{\alpha})$  by sequentially testing all candidate codewords and selecting the one that maximizes the total wideband received power, i.e.,

$$(\hat{\theta}, \hat{\alpha}) = \arg \max_{(\theta_n, \alpha_n^{S_n})} \sum_{m=1}^M \left| \sqrt{P_t} \mathbf{g}_m^H c_m(\theta_n, \alpha_n^{S_n}) + v_m \right|^2. \quad (9)$$

However, since the 2D polar-domain codebook grows with both the angular and distance grid resolutions, exhaustive sweeping leads to prohibitive training overhead [15]. Therefore, exhaustive codebook search is not suitable for practical wideband XL-MIMO near-field beam training.

Recent distance-dependent beam split (DDBS) methods [23] substantially reduce the required pilot overhead by exploiting frequency-dependent beamforming to probe multiple angles and distances simultaneously. However, existing DDBS designs typically assume one dedicated TTD unit per antenna element. In 6G XL-MIMO systems, setting  $N_{\text{TTD}} = N_t$  results in excessive hardware cost, power consumption, and implementation complexity.



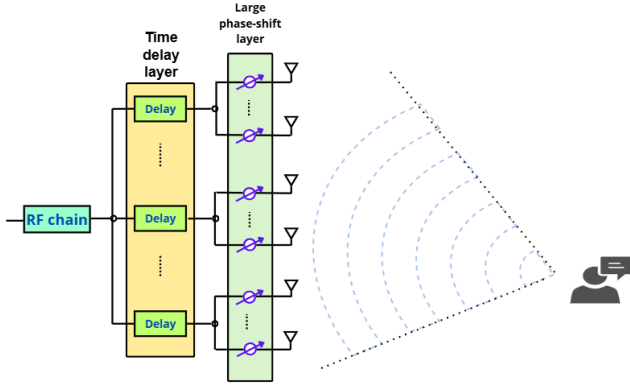


FIGURE 1. Grouped TTD-PS (TTD + phase-shifter) analog precoding architecture.

Motivated by this limitation, this work aims to preserve the fast joint angle–distance probing capability of DDBS while operating under a hardware-constrained AoSA architecture. Specifically, we consider grouped TTD beamforming with  $N_{\text{TTD}} = G = N_t/L$ , where one TTD unit is shared by each sub-array of  $L$  antennas. Under this grouped-delay constraint, our goal is to design a sequence of  $K$  pilot beam configurations  $\{\mathbf{v}_m^{(k)}\}_{k=1}^K$  such that the target search region  $\mathcal{S}$  is covered in the angle–distance domain under the grouped-delay constraint while maintaining a practical pilot budget. Based on the resulting pilot observations, the beam training task is to estimate  $(\hat{\theta}, \hat{\alpha}) \in \mathcal{S}$  and obtain accurate beam alignment without exhaustive 2D sweeping.

### C. PROPOSED SUB-ARRAY-SHARED TTD-PS ARCHITECTURE

As shown in Fig. 1, an AoSA grouped TTD-PS analog precoding architecture is employed to enable subcarrier-dependent near-field focusing via a frequency-dependent delay response and a frequency-flat phase response. In a conventional per-antenna TTD-PS implementation, the  $m$ -th subcarrier weight admits  $[\mathbf{v}_m]_{\hat{n}} \propto e^{-j2\pi f_m \tau_{\hat{n}}} e^{jk_m \hat{\phi}_{\hat{n}}}$ , which requires one tunable delay per antenna and is costly to scale in XL-MIMO. Therefore, we share one TTD within each sub-array while retaining per-antenna phase shifters.

Let  $L$  denote the number of antennas in each sub-array. It is assumed that  $L$  divides  $N_t$ , and we define the number of sub-arrays as  $G \triangleq N_t/L$ . For implementation, the sub-arrays and antennas within each sub-array can be indexed by  $n_s \in \{0, \dots, G-1\}$  and  $\hat{i} \in \{0, \dots, L-1\}$ , respectively, so that the physical antenna index is  $n = n_s L + \hat{i}$ , with  $n \in \{0, \dots, N_t-1\}$ . For analysis, centered (symmetric) indices are used by shifting the origin to the array center:  $\hat{n}_s = n_s - \frac{G-1}{2}$  and  $i = \hat{i} - \frac{L-1}{2}$ . This yields the sub-array (TTD) index  $\hat{n}_s$  and the intra-sub-array index  $i$

$$\hat{n}_s \in \left\{ -\frac{G-1}{2}, \dots, \frac{G-1}{2} \right\},$$

$$i \in \mathcal{I} \triangleq \left\{ -\frac{L-1}{2}, \dots, \frac{L-1}{2} \right\}. \quad (10)$$

Thus, each centered antenna index decomposes as  $\hat{n} = \hat{n}_s L + i$ ,  $i \in \mathcal{I}$ . Based on this index decomposition, we next specify the proposed AoSA TTD-PS analog precoder in three steps: 1) sub-array-shared TTD weights, 2) per-antenna PS weights, and 3) the resulting per-subcarrier beamforming vector.

#### 1) SUB-ARRAY-SHARED TTD WEIGHTS

A *true-time-delay* (TTD) unit realizes a frequency-dependent phase response by *physically delaying* the RF signal. Specifically, applying a delay  $\tau$  produces a phase rotation  $e^{-j2\pi f_m \tau}$  at frequency  $f_m$ , which enables subcarrier-dependent beam control.

As illustrated in Fig. 1, an AoSA TTD-PS architecture is adopted, where *one tunable TTD is shared by all antennas within each sub-array*, while *each antenna is equipped with an individual phase shifter*. Accordingly, for the sub-array indexed by  $\hat{n}_s$ , the TTD beamforming vector can be expressed as

$$[\mathbf{v}_m^{\text{TTD}}]_{\hat{n}_s} = \frac{1}{\sqrt{N_t}} e^{-j2\pi f_m \tau'(\hat{n}_s)} = \frac{1}{\sqrt{N_t}} e^{-jk_m r'(\hat{n}_s)}, \quad (11)$$

where  $\tau'(\hat{n}_s)$  denotes the adjustable delay applied by the shared TTD of sub-array  $\hat{n}_s$  and  $r'(\hat{n}_s) = c \tau'(\hat{n}_s)$  with  $c$  being the speed of light.

To emulate a polar-domain phase profile under sub-array sharing, we parameterize  $r'(\hat{n}_s)$  over the *sub-array-center lattice* (effective spacing  $Ld$ ) using two *design (control) variables*  $(\theta'_t, \alpha'_t)$  as

$$r'(\hat{n}_s) = \hat{n}_s L d \theta'_t - \hat{n}_s^2 L^2 d^2 \alpha'_t, \quad (12)$$

where  $(\theta'_t, \alpha'_t)$  are *TTD control parameters* defined over the sub-array indices, whereas  $(\theta, \alpha)$  characterize the UE location (channel) parameters introduced in (7). Substituting (12) into (11) yields the following expression of the TTD beamforming weight  $\mathbf{v}_m^{\text{TTD}} \in \mathbb{C}^{G \times 1}$

$$[\mathbf{v}_m^{\text{TTD}}(\theta'_t, \alpha'_t)]_{\hat{n}_s} = \frac{1}{\sqrt{N_t}} \exp\left(-jk_m(\hat{n}_s L d \theta'_t - \hat{n}_s^2 L^2 d^2 \alpha'_t)\right). \quad (13)$$

#### 2) PER-ANTENNA PS WEIGHTS

Following the standard TTD-PS/JPTA modeling in [23], the frequency-flat PS network is designed at the center frequency with  $k_c = 2\pi f_c/c$ . The  $\hat{n}$ -th element of the PS beamforming vector can be expressed as

$$[\mathbf{v}_m^{\text{PS}}(\theta'_p, \alpha'_p)]_{\hat{n}} = \frac{1}{\sqrt{N_t}} \exp(-jk_c(\hat{n} d \theta'_p - \hat{n}^2 d^2 \alpha'_p)), \quad (14)$$

where  $\theta'_p$  and  $\alpha'_p$  are the design (control) parameters of the PS network, which shapes the per-antenna phase profile of the frequency-flat PS network.

### 3) OVERALL PER-SUBCARRIER BEAMFORMING VECTOR

Combining the sub-array-level TTD weight and the per-antenna PS weight in (13)–(14), the  $\hat{n}$ -th element of the beamforming vector on subcarrier  $m$  can be written as

$$[\mathbf{v}_m(\theta'_t, \alpha'_t, \theta'_p, \alpha'_p)]_{\hat{n}} = \frac{1}{N_t} \exp(j\phi_m^{\text{TTD}}(g(\hat{n}))) \times \exp(j\phi_m^{\text{PS}}(\hat{n})), \quad (15)$$

where  $g(\hat{n})$  maps antenna index  $\hat{n}$  to its sub-array index, i.e.,  $g(\hat{n}) = \hat{n}_s \triangleq [\hat{n}/L]$ , and  $\phi_m^{\text{TTD}}(\cdot)$  and the phase terms are defined as  $\phi_m^{\text{TTD}}(g(\hat{n})) \triangleq -k_m(g(\hat{n})Ld\theta'_t - g(\hat{n})^2L^2d^2\alpha'_t)$  and  $\phi_m^{\text{PS}}(\hat{n}) \triangleq -k_c(\hat{n}d\theta'_p - \hat{n}^2d^2\alpha'_p)$ .

To simplify subsequent analysis, the AoSA TTD–PS beamforming vector is next expressed in a compact vector form. Recall that  $N_t = GL$  antennas are partitioned into  $G$  sub-arrays of size  $L$ . Let  $\mathbf{v}_m^{\text{TTD}} \in \mathbb{C}^{G \times 1}$  collect the sub-array TTD coefficients and define the fixed replication matrix  $\mathbf{R} \triangleq \mathbf{I}_G \otimes \mathbf{1}_L \in \{0, 1\}^{N_t \times G}$ , so that the expanded antenna-domain TTD vector is  $\mathbf{r}_m = \mathbf{R}\mathbf{v}_m^{\text{TTD}} \in \mathbb{C}^{N_t \times 1}$ . Let  $\mathbf{v}^{\text{PS}} \in \mathbb{C}^{N_t \times 1}$  collect the element-wise PS coefficients. Then the per-subcarrier analog beamforming vector is

$$\mathbf{v}_m = \frac{1}{N_t} (\mathbf{v}^{\text{PS}} \odot \mathbf{r}_m), \quad (16)$$

where  $\odot$  denotes the Hadamard product.

*Remark 1 (Extension to Planar Arrays: A UPA-Based Sketch):* Although the proposed framework is developed for a ULA in order to keep the wideband grouped-TTD analysis tractable, the same design philosophy can be extended to more practical planar-array configurations. To make this extension concrete, consider a uniform planar array (UPA) deployed on the  $x$ - $z$  plane, where the antenna element indexed by  $(n_x, n_z)$  is located at

$$\mathbf{p}_{n_x, n_z} = (n_x d_x, 0, n_z d_z), \quad (17)$$

with  $n_x \in \{-N_x, \dots, N_x\}$ ,  $n_z \in \{-N_z, \dots, N_z\}$ , and

$$N_t = (2N_x + 1)(2N_z + 1). \quad (18)$$

Let  $(\theta_x, \theta_z)$  denote two normalized spatial-direction variables and let  $r$  denote the user range. Under a second-order near-field approximation, and by neglecting the mixed bilinear term for an illustrative separable extension, the path-length difference can be written as

$$r_{n_x, n_z} - r \approx -n_x d_x \theta_x - n_z d_z \theta_z + \alpha_x (n_x d_x)^2 + \alpha_z (n_z d_z)^2, \quad (19)$$

where  $\alpha_x$  and  $\alpha_z$  are effective curvature-related parameters along the two spatial dimensions. Accordingly, the UPA near-field steering response at subcarrier  $m$  can be written in the approximate form

$$[\mathbf{c}_m^{\text{UPA}}(\theta_x, \theta_z, \alpha_x, \alpha_z)]_{n_x, n_z} = \frac{1}{\sqrt{N_t}} \exp(jk_m(n_x d_x \theta_x + n_z d_z \theta_z - \alpha_x (n_x d_x)^2 - \alpha_z (n_z d_z)^2)), \quad (20)$$

which admits an approximately separable structure across the two spatial dimensions.

For a 2D AoSA implementation, let each tile contain  $L_x \times L_z$  antennas and write

$$n_x = g_x L_x + i_x, \quad n_z = g_z L_z + i_z, \quad (21)$$

where  $(g_x, g_z)$  denotes the subarray index and  $(i_x, i_z)$  the intra-subarray index. Then the grouped-TTD control can be defined on the intra-subarray-center grid as

$$r'_t(g_x, g_z) = g_x L_x d_x \theta'_{t,x} + g_z L_z d_z \theta'_{t,z} - \alpha'_{t,x} (g_x L_x d_x)^2 - \alpha'_{t,z} (g_z L_z d_z)^2, \quad (22)$$

with the corresponding sub-array-level TTD weight given by

$$[\mathbf{v}_m^{\text{TTD}}]_{g_x, g_z} = \frac{1}{\sqrt{N_t}} \exp(-jk_m r'_t(g_x, g_z)). \quad (23)$$

Based on (19)–(23), the LS focusing idea can be generalized by minimizing the residual phase mismatch over both spatial dimensions, thereby obtaining an effective focusing point in azimuth–elevation–range space. Under planar arrays, the strong-gain region becomes a two-dimensional angular sector rather than a one-dimensional angular sector. Consequently, full coverage requires sector shifting across both spatial dimensions together with distance interleaving, and therefore the pilot overhead is expected to increase relative to the ULA case. To keep the paper focused and analytically tractable, the full derivations and simulations remain based on the ULA baseline, while the added UPA discussion is intended to demonstrate the technical extensibility of the proposed grouped-delay design rather than to provide a complete UPA performance evaluation.

### III. ARRAY GAIN ANALYSIS UNDER SUB-ARRAY TTD–PS PRECODING

In this section, the beampattern (array gain) produced by the proposed proposed AoSA TTD–PS architecture is analyzed and connected to the near-field polar-domain location parameters  $(\theta, \alpha)$ . Our goal is to derive a closed-form expression for the array gain, which clarifies how delay sharing within each sub-array affects wideband near-field beam training.

#### A. ARRAY-GAIN DEFINITION

The analog beamforming vector at subcarrier  $m$  consists of a *sub-array-shared* TTD layer and a *per-antenna* PS layer, whose weights are given in (13)–(14). The near-field polar steering vector  $\mathbf{c}_m(\theta, \alpha)$  is defined in (7). Using the AoSA mapping and the element-wise expression of the beamforming vector in (16), we measure the alignment for a user located at  $(\theta, \alpha)$  via the normalized array gain

$$G_m(\theta, \alpha) = |\mathbf{v}_m^H \mathbf{c}_m(\theta, \alpha)|. \quad (24)$$

By expanding (24) under the AoSA mapping in (15), the following two-level summation representation is obtained.

$$G_m(\theta, \alpha) = \frac{1}{N_t} \left| \sum_{\hat{n}_s=-Q}^Q e^{-jk_m(\hat{n}_s L d \theta'_t - \hat{n}_s^2 L^2 d^2 \alpha'_t)} \times \sum_{i \in \mathcal{I}} e^{j((\hat{n}_s L + i)d x_m - (\hat{n}_s L + i)^2 d^2 y_m)} \right|. \quad (25)$$

where  $\mathcal{I}$  is defined in (10). Define the residual intra-sub-array phase-mismatch terms

$$x_m \triangleq k_m \theta - k_c \theta'_p \quad \text{and} \quad y_m \triangleq k_m \alpha - k_c \alpha'_p, \quad (26)$$

where  $x_m$  and  $y_m$  represent the residual *linear* (angle-related) and *quadratic* (curvature/range-related) phase coefficients between the channel parameters  $(\theta, \alpha)$  and the PS design parameters  $(\theta'_p, \alpha'_p)$ .

Equation (25) reveals a two-scale structure arising from TTD sharing within each sub-array. The outer summation over  $\hat{n}_s$  captures the phase progression across sub-array centers (effective spacing  $Ld$ ) governed by the shared TTD layer, while the inner summation over  $i \in \mathcal{I}$  captures the coherent combining among the  $L$  antennas within each sub-array governed by the mismatch pair  $(x_m, y_m)$  in (26). When the residual phase mismatches  $x_m$  and  $y_m$  are sufficiently small across the elements of each sub-array, the intra-sub-array phase variation is small and the array gain is large; otherwise, the phase dispersion across the  $L$  antennas causes partial cancellation and reduced gain.

### B. LEAST-SQUARES (LS) FOCUSING WITH SHARED TTD

In the conventional (non-sub-arrayed) TTD-PS architecture ( $L = 1$ ), the available independent phase-control variables are sufficient to match the desired near-field polar phase profile across all antenna elements at each subcarrier. In contrast, under the proposed sub-array-shared TTD architecture ( $L > 1$ ), a single delay is replicated over  $L$  antennas, which generally makes it impossible to achieve perfect phase matching across the entire aperture. As a result, we propose an LS method to find effective focus  $(\theta_m, \alpha_m)$  that minimizes the residual phase mismatch over the antenna elements. Consequently, for a fixed parameter quadruple  $(\theta'_t, \alpha'_t, \theta'_p, \alpha'_p)$ , the beamforming vector on subcarrier  $m$  is best interpreted as *approximately* focusing at the corresponding effective location  $(\theta_m, \alpha_m)$ .

#### 1) RESIDUAL PHASE MISMATCH

Consider attempting to focus on subcarrier  $m$  at an effective polar location  $(\theta_m, \alpha_m)$ . For antenna index  $\hat{n}$  (belonging to sub-array  $g(\hat{n})$ ), the residual phase mismatch (in radians) that quantifies the phase error at antenna  $\hat{n}$  between the phase profile implemented by the proposed TTD-PS precoder (i.e., the combined contribution of the sub-array-shared TTD term and the per-antenna PS term) and the ideal polar-domain phase that would focus the beam at the effective location  $(\theta_m, \alpha_m)$  on subcarrier  $m$

$$\begin{aligned} \varphi(\hat{n}) \triangleq & k_m \left( g(\hat{n}) L d \theta'_t - g(\hat{n})^2 L^2 d^2 \alpha'_t \right) \\ & + k_c \left( \hat{n} d \theta'_p - \hat{n}^2 d^2 \alpha'_p \right) \\ & - k_m \left( \hat{n} d \theta_m - \hat{n}^2 d^2 \alpha_m \right), \end{aligned} \quad (27)$$

where  $k_c = 2\pi f_c / c$ , and the sub-array index mapping is  $g(\hat{n}) \triangleq \lfloor \hat{n} / L \rfloor$  with  $\hat{n} = \hat{n}_s L + i$ ,  $i \in \mathcal{I}$ .

Accordingly, perfect focusing is achieved when  $\varphi(\hat{n}) = 0$  for all  $\hat{n}$ , i.e., all the antenna phases are fully aligned. Otherwise,  $\varphi(\hat{n}) \neq 0$  quantifies the residual phase misalignment, and the resulting array-gain loss stems from the sub-array-shared delay constraint.

#### 2) LS FOCUSING OBJECTIVE

The effective focusing point  $(\theta_m, \alpha_m)$  is defined by minimizing the least-squares residual phase error over the aperture  $J(\theta_m, \alpha_m) \triangleq \sum_{\hat{n}=-N}^N (\varphi(\hat{n}))^2$ . Since  $\varphi(\hat{n})$  is affine in  $(\theta_m, \alpha_m)$ , the cost  $J(\theta_m, \alpha_m)$  is a convex quadratic function and admits a unique global minimizer. The first-order optimality conditions for the LS objective reduce to the following normal equations:

$$\sum_{\hat{n}=-N}^N \varphi(\hat{n}) \hat{n} = 0, \quad \sum_{\hat{n}=-N}^N \varphi(\hat{n}) \hat{n}^2 = 0. \quad (28)$$

#### 3) CLOSED-FORM LS FOCUSING SOLUTION

Solving (28) gives the following closed-form mapping from the hardware parameters  $(\theta'_t, \alpha'_t, \theta'_p, \alpha'_p)$  to the effective focus point  $(\theta_m, \alpha_m)$  on subcarrier  $m$ :

$$\theta_m = \frac{k_c}{k_m} \theta'_p + \rho_\theta \theta'_t, \quad (29)$$

$$\alpha_m = \frac{k_c}{k_m} \alpha'_p + \rho_\alpha \alpha'_t, \quad (30)$$

where the architecture-dependent coefficients are

$$\rho_\theta \triangleq \frac{L C_1}{S_2}, \quad \rho_\alpha \triangleq \frac{L^2 C_2}{S_4}, \quad (31)$$

with the symmetry sums  $S_2 \triangleq \sum_{\hat{n}=-N}^N \hat{n}^2$ ,  $S_4 \triangleq \sum_{\hat{n}=-N}^N \hat{n}^4$ ,  $C_1 \triangleq \sum_{\hat{n}=-N}^N \hat{n} \hat{n}_s$ ,  $C_2 \triangleq \sum_{\hat{n}=-N}^N \hat{n}^2 \hat{n}_s^2$ , and  $\hat{n}_s = g(\hat{n})$  is the sub-array index of element  $\hat{n}$ . Note that centered (symmetric) indexing is adopted, i.e.,  $\hat{n} \in \{-N, \dots, N\}$  and  $\hat{n}_s \in \{-Q, \dots, Q\}$  with  $i \in \mathcal{I}$  symmetric around zero. Hence, by symmetry, all odd-order sums vanish, e.g.,  $\sum_{\hat{n}=-N}^N \hat{n} = 0$ ,  $\sum_{\hat{n}=-N}^N \hat{n}^3 = 0$ ,  $\sum_{i \in \mathcal{I}} i = 0$ ,  $\sum_{i \in \mathcal{I}} i^3 = 0$ , and similarly  $\sum_{\hat{n}_s=-Q}^Q \hat{n}_s = 0$  and  $\sum_{\hat{n}_s=-Q}^Q \hat{n}_s^3 = 0$ . These properties will be used to simplify the LS normal equations and obtain closed-form expressions. In the special case  $L = 1$  (non-sub-arrayed,  $\hat{n}_s = \hat{n}$ , which implies  $C_1 = S_2$  and  $C_2 = S_4$ , hence  $\rho_\theta = \rho_\alpha = 1$ ), (29)–(30) reduce to the conventional (non-sub-arrayed) focusing relation. For contiguous symmetric sub-arrays, the symmetry sums  $S_2$ ,  $S_4$ ,  $C_1$ , and  $C_2$  admit closed-form expressions, which are provided in Appendix A for completeness.

Note that  $\rho_\theta$  and  $\rho_\alpha$  in (31) depend only on  $(N, L)$  and can be precomputed once for a given array configuration. These LS-induced effective focusing points  $(\theta_m, \alpha_m)$  are used in Section IV-C to characterize the subcarrier-dependent array gain and the resulting beam-split behavior under sub-array TTD-PS.

#### IV. SUB-ARRAY TTD-PS BEAM TRAINING: SECTORING AND PERIODIC FOCUSING

In this section, the array-gain expression derived in the previous section is interpreted and the key structural properties induced by sub-array-shared TTDs are revealed. Specifically, the mapping  $\hat{n} = \hat{n}_s L + i$  decomposes the beamforming gain in (25) into an outer (sub-array-lattice) response and an inner (intra-sub-array) response, which naturally leads to sectoring behavior. Moreover, coherent focusing occurs when the relevant phase terms align modulo  $2\pi$ ; hence multiple focusing solutions exist and can be indexed by integers. As the subcarrier frequency varies, these integer-indexed solutions shift, so the focusing points trace strip-like trajectories in the angle–range domain. These properties will be leveraged in the next section to design an efficient beam training scheme.

##### A. TWO-SCALE GAIN AND SECTORING IN THE FAR-FIELD (FOR INTUITION)

To highlight the impact of sub-array-shared TTDs on the beampattern, we first study the far-field special case by setting  $\alpha = 0$ . Starting from the normalized array gain in (25), substituting  $\alpha = 0$  and using the sub-array decomposition  $\hat{n} = \hat{n}_s L + i$  with  $\hat{n}_s \in \{-Q, \dots, Q\}$  and  $i \in \mathcal{I}$ , the gain at subcarrier  $m$  can be reorganized into an outer sum across sub-arrays and an inner sum within each sub-array.

**Proposition 1:** [Two-scale factorization] For  $\alpha = 0$ , the normalized sub-array TTD-PS gain at subcarrier  $m$  decomposes into two Dirichlet-type kernels corresponding to inter-sub-array and intra-sub-array combining:

$$G_m(\theta, 0) = \frac{1}{N_t} \left| \sum_{\hat{n}_s=-Q}^Q e^{j\hat{n}_s \Omega_m(\theta)} \right| \left| \sum_{i \in \mathcal{I}} e^{ji z_m(\theta)} \right|, \quad (32)$$

where

$$\Omega_m(\theta) \triangleq Ld(k_m(\theta - \theta'_t) - k_c \theta'_p), \quad z_m(\theta) \triangleq d(k_m \theta - k_c \theta'_p). \quad (33)$$

Proposition 1 reveals a clear two-scale structure as a product of an inner and an outer factor: the outer factor is governed by the sub-array lattice (effective spacing  $Ld$ ), while the inner factor is governed by the intra-sub-array combining across the  $L$  antennas.

*Proof:* (See Appendix B). ■

To characterize the candidate angular maxima created by the outer factor, define  $D_G(\Omega_m(\theta)) \triangleq \sum_{\hat{n}_s=-Q}^Q e^{j\hat{n}_s \Omega_m(\theta)}$ . Since  $D_G(\cdot)$  is  $2\pi$ -periodic,  $|D_G(\Omega_m(\theta))|$  is maximized when  $\Omega_m(\theta) = 2\pi p$ ,  $p \in \mathbb{Z}$ . Substituting the periodicity in (33) yields the family of candidate peak angles

$$\theta_{m,p} = \theta'_t + \frac{f_c}{f_m} \theta'_p + \frac{\lambda_m}{Ld} p, \quad p \in \mathbb{Z}, \quad (34)$$

with angular spacing

$$\Delta\theta_{m,p} = \frac{\lambda_m}{Ld}, \quad (35)$$

where  $\lambda_m = c/f_m$  is the wavelength at subcarrier  $m$ . Thus, the sub-array lattice induces a periodic alias family of candidate maxima indexed by  $p$ , and the period depends on  $Ld$  rather than  $d$ , reflecting the coarser sub-array sampling.

The inner factor assigns an amplitude weight to each candidate peak. At  $\theta = \theta_{m,p}$ , using  $k_m \frac{f_c}{f_m} = k_c$  and  $k_m \lambda_m = 2\pi$ , the intra-sub-array phase is

$$\delta_{m,p} = z_m(\theta_{m,p}) = k_m d \theta'_t + \frac{2\pi}{L} p. \quad (36)$$

Hence, the normalized intra-sub-array factor at the  $p$ -th candidate peak is

$$\left| \frac{1}{L} \sum_{i \in \mathcal{I}} e^{ji z_m(\theta_{m,p})} \right| = \left| \frac{\sin(\frac{L}{2} \delta_{m,p})}{L \sin(\frac{1}{2} \delta_{m,p})} \right|. \quad (37)$$

Equations (34)–(35) show that the outer factor generates a periodic family of candidate peaks indexed by  $p$ . Equation (36) shows that the amplitude of each candidate peak is weighted by the intra-sub-array Dirichlet kernel evaluated at  $\delta_{m,p}$ , which advances in steps of  $2\pi/L$  as  $p$  increases. As  $L$  grows, the intra-sub-array kernel becomes more selective, so only candidates within a limited angular region remain strong, forming angular sectors.

**Remark 2 (Consistency with the Ungrouped Case):** When  $L = 1$ ,  $\mathcal{I} = \{0\}$  and the intra-sub-array factor in the array-gain expression reduces to one. Therefore, the masking (sectoring) effect vanishes, recovering the conventional per-antenna (ungrouped) behavior.

##### B. NEAR-FIELD PHASOR ALIGNMENT AND PERIODIC FOCUSING PRINCIPLE (GENERAL CASE, $\alpha \neq 0$ )

Under the sub-array TTD-PS beamforming vector, the normalized array gain at subcarrier  $m$  follows (25). Since (25) contains a residual intra-sub-array phase mismatch term, perfect focusing (unit gain) is not achievable in general when  $L > 1$ . This motivates the LS-based focusing design in Subsection III-B, which minimizes the residual mismatch and improves the peak gain.

Compared with the far-field case, the additional quadratic term involving  $y_m$  couples the range (curvature) parameter  $\alpha$  into the phase progression across the array.

To obtain the coherent focusing locations, we adopt a *phasor-alignment* interpretation of the outer sum in (25). After substituting  $\hat{n} = \hat{n}_s L + i$  and collecting the terms that depend only on the sub-array index  $\hat{n}_s$ , the outer summation can be viewed as a sum of unit-modulus phasors  $e^{j\phi_s}$  indexed by  $\hat{n}_s$ , with

$$\phi_g = Ld \hat{n}_s x'_m - L^2 d^2 \hat{n}_s^2 y'_m. \quad (38)$$

Here the sub-array-level residuals are  $x'_m \triangleq x_m - k_m \theta'_t$  and  $y'_m \triangleq y_m - k_m \alpha'_t$ . The outer sum is maximized when these phasors point in the same direction, i.e., when  $\phi_g$  is constant modulo  $2\pi$  for all  $\hat{n}_s$ . A sufficient way to enforce this condition is to align the linear and quadratic terms over the sub-array lattice modulo  $2\pi$ , namely  $Ld x'_m = 2\pi p_m$  and  $L^2 d^2 y'_m = 2\pi q_m$  for some integers  $p_m, q_m \in \mathbb{Z}$ . This modulo- $2\pi$  alignment will



lead to the periodic focusing families in the polar domain derived next.

**Proposition 2:** [Sub-array periodic focusing principles in the polar domain] Under the sub-array TTD-PS architecture, the coherent near-field focusing points  $(\theta_m, \alpha_m)$  on subcarrier  $m$  form periodic families parameterized by integers  $(p_m, q_m)$ :

$$\theta_m = \rho_\theta \theta'_t + \frac{f_c}{f_m} \theta'_p + \frac{\lambda_m}{Ld} p_m, \quad (39)$$

$$\alpha_m = \rho_\alpha \alpha'_t + \frac{f_c}{f_m} \alpha'_p + \frac{\lambda_m}{L^2 d^2} q_m, \quad (40)$$

where  $\rho_\theta$  and  $\rho_\alpha$  are given in (31) and capture the effective scaling induced by sub-array sharing (as in the LS focusing result).

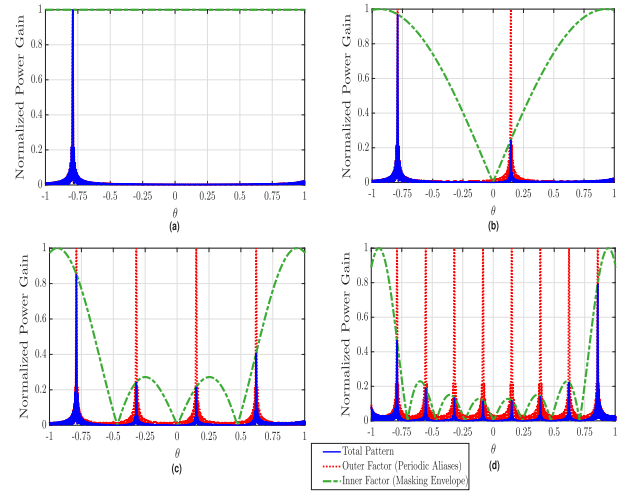
*Proof:* (See Appendix C). ■

**Connection back to the far-field case (Proposition 1):** Setting  $y'_m = 0$  removes the curvature term and reduces (38) to the outer angular periodicity, thereby recovering the far-field candidate-peak structure. Moreover, (39)–(40) show that the coherent focusing points  $(\theta_m, \alpha_m)$  depend on the subcarrier frequency  $f_m$ . Hence, as the subcarrier index  $m$  sweeps across the OFDM band, the focusing locations move continuously and trace inclined trajectories in the  $(\theta, \alpha)$  plane. We refer to these trajectories as *strips*, and to the corresponding frequency-dependent coverage mechanism as *strip scanning*. The integers  $p_m$  and  $q_m$  select the angular- and curvature-alias families, respectively, while the frequency dependence determines how each family is scanned across subcarriers. Due to the bounded angular domain  $\theta \in [-1, 1]$ , some strip trajectories may approach  $\theta = \pm 1$  and wrap around, which can create larger gaps between neighboring strips near the boundary. This observation motivates the coverage-based pilot design developed later in Section V, where the one-pilot band-wide coverage is quantified and the pilot parameters are chosen to avoid holes.

### C. SECTORING VISUALIZATION: BEAM PATTERNS FOR DIFFERENT SUB-ARRAY SIZES

Equation (32) indicates that the far-field beampattern under sub-array TTD-PS naturally decomposes into an outer sub-array-lattice term and an inner intra-sub-array term. The outer term (with effective spacing  $Ld$ ) generates periodic candidate peaks across the angular domain, while the inner term acts as a coherence “mask” that preserves only those candidates whose intra-sub-array phases remain aligned. As the sub-array size  $L$  increases, this masking becomes more selective, which narrows the angular region over which strong gain can be maintained and therefore strengthens the sectoring behavior.

Fig. 2 illustrates this effect by plotting normalized beampatterns for  $L = 1, 2, 4, 8$ . When  $L$  is small, the masking is weaker and multiple replicated peaks remain visible, resulting in a wider strong-gain region. In contrast, for larger  $L$ , most replicas are suppressed and only a small number of dominant peaks remain, producing a narrower strong-gain sector. This



**FIGURE 2.** Normalized beampattern decomposition of the TTD-PS sub-array architecture where  $f_m = 32.50$  GHz,  $f_c = 30$  GHz,  $\theta'_t = -31.73$ ,  $\theta'_p = 31$ , for different sub-array sizes: (a)  $L = 1$ , (b)  $L = 2$ , (c)  $L = 4$ , (d)  $L = 8$ .

visualization is consistent with the analysis later in Section V, where the one-pilot strong-gain sector width is shown to decrease as  $L$  increases.

### D. TRAINING DESIGN IMPLICATIONS OF PERIODIC FOCUSING

The periodic focusing principles (39)–(40) indicate that, for the proposed sub-array TTD-PS architecture, the coherent focusing locations  $(\theta_m, \alpha_m)$  vary with the subcarrier frequency and sweep along inclined trajectories in the  $(\theta, \alpha)$  plane. Consequently, a single pilot (with fixed  $(\theta'_t, \theta'_p, \alpha'_t, \alpha'_p)$ ) can simultaneously probe multiple angle–distance hypotheses across subcarriers via a multi-strip scanning pattern. However, sub-array/grouping introduces an intra-sub-array coherence (masking) effect that concentrates strong gain into a limited angular range/sector for a fixed  $(\theta'_t, \theta'_p)$ , and the resulting strips may leave distance-domain coverage holes between adjacent trajectories. These observations motivate the proposed practical training design: (i) sector shifting across pilots to cover the full angular range, and (ii) interleaved distance pilots within each sector to fill uncovered distance gaps.

## V. PROPOSED SUB-ARRAY DISTANCE-DEPENDENT BEAM-SPLIT BEAM TRAINING

In this section, a practical sub-array-shared TTD-PS beam training scheme is developed based on the sectoring and periodic focusing principles derived in Sec. IV. By designing multi-strip probing patterns across subcarriers, the proposed method jointly searches the angle–range (polar domain) without exhaustive 2D scanning. The sector-shift and distance-interleaving rules are further provided to ensure full coverage with a reasonable number of pilots.

### A. SINGLE-PILOT STRONG-GAIN SECTOR WIDTH UNDER SUB-ARRAY SHARING

To design a multi-pilot training scheme, we first characterize the architectural limitation of AoSA beamforming, the

angular sector over which a single pilot maintains strong array gain, termed the *single-pilot strong-gain sector width*. This width is obtained by first determining the interval of the intra-sub-array phase variable  $\delta$  in (37) over which the intra-sub-array gain factor in (32) exceeds the 3-dB threshold, and then converting this  $\delta$ -interval into an angular width using the mapping between  $\delta_{m,p}$  and  $\theta_{m,p}$  in (36). We now examine the intra-sub-array factor. From Proposition 1, the far-field array gain for a sub-array of size  $L$  factorizes into an inter-sub-array term and an intra-sub-array term. At the peaks of the inter-sub-array term, the overall array gain is governed by the intra-sub-array response, which can be expressed as the Dirichlet kernel

$$C_L(\delta) = \left| \frac{\sin(L\delta/2)}{L \sin(\delta/2)} \right|. \quad (41)$$

The kernel satisfies  $C_L(\delta) = 1$  at  $\delta = 2\pi n$  for any integer  $n$ . Under the standard 3-dB (half-power) criterion, strong coverage is declared when  $|C_L(\delta)| \geq 1/\sqrt{2}$ . To characterize the corresponding neighborhood around each peak, let  $\delta = 2\pi n + \epsilon$ , where  $\epsilon$  denotes the deviation from the peak. Utilizing the periodicity and symmetry of the sine function, this yields

$$C_L(2\pi n + \epsilon) = \left| \frac{\sin(L\epsilon/2)}{L \sin(\epsilon/2)} \right|. \quad (42)$$

For the mainlobe,  $\epsilon$  is small, so the small-angle approximation  $\sin(\epsilon/2) \approx \epsilon/2$  applies, which yields

$$C_L(2\pi n + \epsilon) \approx \left| \frac{\sin(L\epsilon/2)}{L \cdot (\epsilon/2)} \right| = \left| \frac{\sin(u)}{u} \right|, \quad (43)$$

where  $u \triangleq L\epsilon/2$ . The 3-dB condition  $\left| \frac{\sin(u_{3dB})}{u_{3dB}} \right| = \frac{1}{\sqrt{2}}$  is solved numerically, yielding the smallest positive solution  $u_{3dB} \approx 1.39$ . Translating back to  $\epsilon$  gives  $|L\epsilon/2| \leq u_{3dB}$ , i.e.,  $|\epsilon| \leq 2u_{3dB}/L \approx 2.78/L$ . The corresponding two-sided interval in  $\delta$  is

$$\Delta\delta_{\text{strong}} = \frac{5.56}{L}. \quad (44)$$

This result shows that the  $\delta$ -range over which the intra-sub-array factor provides strong gain scales as  $O(1/L)$ , which explains the sectoring effect observed in Fig. 2.

With the  $\delta$ -range  $\Delta\delta_{\text{strong}}$  in (44), the corresponding angular width follows from the local mapping between  $\delta$  and  $\theta_{m,p}$ . For a sufficiently small interval,  $\Delta\delta \approx \left| \frac{d\delta}{d\theta_{m,p}} \right| \cdot \Delta\theta_{m,p}$ , which yields

$$\Delta\theta_{m,p} = \frac{\Delta\delta}{\left| \frac{d\delta}{d\theta_{m,p}} \right|}. \quad (45)$$

The derivative  $|d\delta/d\theta_{m,p}|$  quantifies the sensitivity of  $\delta$  to variations in  $\theta_{m,p}$ , a larger value implies that a fixed  $\delta$ -budget maps to a narrower angular sector.

Evaluating  $|d\delta/d\theta_{m,p}|$  requires the relationship between  $\delta$  and  $\theta_{m,p}$ . From the periodic focusing principle in Section IV, for a fixed alias branch indexed by  $p$ , the inner phase at the

focused direction follows from (36). Using  $k_md = \pi f_m/f_c = \pi\xi_m$  with  $\xi_m \triangleq f_m/f_c$ , we obtain

$$\delta = \pi\xi_m\theta'_t + \frac{2\pi p}{L}. \quad (46)$$

The focused direction is given by (39):

$$\theta_{m,p} = \rho_\theta\theta'_t + \frac{f_c}{f_m} \left( \theta'_p + \frac{2p}{L} \right). \quad (47)$$

For compactness, define the *composite angular parameter*

$$S \triangleq \theta'_p + \frac{2p}{L}. \quad (48)$$

Then the focused direction becomes

$$\theta_{m,p} = \rho_\theta\theta'_t + \frac{S}{\xi_m}. \quad (49)$$

The derivative  $d\delta/d\theta_{m,p}$  follows from the chain rule with respect to the normalized frequency  $\xi_m$ . From (46) and (49), we have  $d\delta/d\xi_m = \pi\theta'_t$  and  $d\theta_{m,p}/d\xi_m = -S/\xi_m^2$ , which yields

$$\frac{d\delta}{d\theta_{m,p}} = \frac{d\delta/d\xi_m}{d\theta_{m,p}/d\xi_m} = -\frac{\pi\theta'_t\xi_m^2}{S}. \quad (50)$$

Consequently,

$$\left| \frac{d\delta}{d\theta_{m,p}} \right| = \frac{\pi|\theta'_t|\xi_m^2}{|S|}. \quad (51)$$

Equation (51) characterizes the sensitivity of  $\delta$  to  $\theta_{m,p}$ . Increasing  $|\theta'_t|$  or  $\xi_m^2$  increases  $|d\delta/d\theta_{m,p}|$  and therefore reduces the angular sector associated with a fixed  $\delta$ -range, whereas a larger  $|S|$  decreases  $|d\delta/d\theta_{m,p}|$  and widens the sector. Equation (51) depends on the subcarrier frequency  $f_m$  through  $\xi_m$ . To guarantee strong gain over the entire band, we adopt a worst-case design by maximizing  $|d\delta/d\theta_{m,p}|$  across subcarriers. Since  $|d\delta/d\theta_{m,p}| \propto \xi_m^2$ , the maximum is attained at the highest frequency  $f_H$ , where  $\xi_H \triangleq f_H/f_c$ . Accordingly,

$$\max_m \left| \frac{d\delta}{d\theta_{m,p}} \right| = \frac{\pi|\theta'_t|\xi_H^2}{|S|}. \quad (52)$$

Substituting  $\Delta\delta_{\text{strong}} = 5.56/L$  from (44) and (52) into the conversion formula (45) yields the single-pilot strong-gain sector width

$$W_{\text{strong}}^{(\text{pilot})} = \frac{5.56|S|}{L\pi|\theta'_t|\xi_H^2}. \quad (53)$$

The number of sector shifts required to cover a target angular span  $\theta_{\text{tot}} \triangleq \theta_{\text{max}} - \theta_{\text{min}}$  is then approximated as

$$N_{\text{sec}} = \left\lceil \frac{\theta_{\text{tot}}}{W_{\text{strong}}^{(\text{pilot})}} \right\rceil. \quad (54)$$

Given  $N_{\text{sec}}$ , sector-dependent parameter pairs  $(\theta_t^{(s)}, \theta_p^{(s)})$  are designed to shift the strong-gain sector across  $[\theta_{\text{min}}, \theta_{\text{max}}]$ , followed by allocating additional pilots within each sector to refine the curvature (distance) coverage.

## B. SECTOR-SHIFT CONFIGURATION OF $(\theta'_t(S), \theta'_p(S))$ (ANGLE-DOMAIN PARAMETERS)

In the non-sub-array TTD-PS architecture, the parameters  $(\theta'_t, \theta'_p)$  can be configured such that the multi-frequency beams sweep the angular domain within each strip, and different pilots mainly shift the strip intercept to fill distance-domain gaps. With sub-array TTD-PS, however, the additional intra-sub-array coherence (masking) factor restricts the strong-gain region over a limited angle span for a fixed  $(\theta'_t, \theta'_p)$ . As a result, increasing the number of pilots improves distance coverage but does not sufficiently shift the angular location of the strong-gain sector. To guarantee uniform strong coverage over the target angular range  $\theta_{\text{tot}}$ , sector shifting is introduced by reconfiguring  $(\theta'_t(s), \theta'_p(s))$  across multiple sector indices.

Based on the single-pilot angular-coverage (sector-width), computed using the 3-dB criterion in (53), the angle-related TTD-PS parameters  $(\theta'_t(s), \theta'_p(s))$  are configured to realize sector shifting across  $N_{\text{sec}}$  sector-shifted pilot configurations. Specifically, after selecting the required number of sectors  $N_{\text{sec}}$  from (54), one parameter pair  $(\theta'_t(s), \theta'_p(s))$  is designed for each sector  $s$ , such that the focused directions  $\{\theta_m\}_{m=1}^M$  sweep continuously within that sector (i.e., without holes), while staying inside  $[-1, 1]$ .

Using (39), for a fixed integer index  $p_m$  (i.e., along one strip), the focused direction on subcarrier  $m$  is

$$\theta_m = \rho_\theta \theta'_t(s) + \frac{f_c}{f_m} \theta'_p(s) + \frac{\lambda_m}{Ld} p_m, \quad p_m \in \mathbb{Z}. \quad (55)$$

Equivalently, using  $\lambda_m = c/f_m$  and  $d = \lambda_c/2 = c/(2f_c)$ , (55) can be rewritten as

$$\theta_m = \rho_\theta \theta'_t(s) + \frac{f_c}{f_m} \left( \theta'_p(s) + \frac{2}{L} p_m \right). \quad (56)$$

For sector design, the alias index is kept fixed along one strip (i.e.,  $p_{m+1} = p_m$ ) and we control the frequency-driven drift across adjacent subcarriers.

Using (56) with  $p_{m+1} = p_m$ , the adjacent-subcarrier drift is

$$|\theta_{m+1} - \theta_m| = \left| f_c \left( \theta'_p(s) + \frac{2}{L} p_m \right) \left( \frac{1}{f_{m+1}} - \frac{1}{f_m} \right) \right|. \quad (57)$$

To ensure that the 3-dB regions of neighboring subcarriers overlap along the strip, the sufficient no-hole condition is imposed as

$$|\theta_{m+1} - \theta_m| \leq \Delta\theta_{3\text{dB}}, \quad (58)$$

where  $\Delta\theta_{3\text{dB}} = \eta_\theta^{f_m} + \eta_\theta^{f_{m+1}}$ , which yields the design inequality

$$\left| \theta'_p(s) + \frac{2}{L} p_m \right| \leq \frac{\Delta\theta_{3\text{dB}}}{f_c \left( \frac{1}{f_{m+1}} - \frac{1}{f_m} \right)}. \quad (59)$$

For improved efficiency, it is beneficial to introduce a safety factor  $\gamma \in [0, 1]$  to ensure that the strip coverage remains continuous across subcarriers. This leads to the following design inequality

$$\left| \theta'_p(s) + \frac{2}{L} p_m \right| \leq \frac{\gamma \Delta\theta_{3\text{dB}}}{f_c \left( \frac{1}{f_{m+1}} - \frac{1}{f_m} \right)}. \quad (60)$$

This constraint is the core requirement because it limits the sweep strength so the strip is scanned densely enough to avoid holes. To obtain a simple closed-form rule valid over the whole band, a conservative worst-case design is adopted by evaluating the adjacent-subcarrier drift at the lowest subcarrier and the 3-dB sector width at the highest subcarrier, for each sector  $s$ ,

$$S_m^{(s)} \triangleq \theta'_p(s) + \frac{2}{L} p_m^{(s)}, \quad (61)$$

where  $p_m^{(s)}$  is the integer associated with the last subcarrier in that sector. Substituting the 3-dB angular width  $\Delta\theta_{3\text{dB}} \approx 1.76 \frac{f_c}{N_t B}$  into the no-hole condition (59) and applying a band-edge worst-case design over the whole band leads to the band-edge approximation,  $S_m^{(s)}$  is bounded as  $S_m^{(s)} \leq \frac{1.76 \gamma f_L M}{N_t B}$ . Accordingly,  $S_m^{(s)}$  is chosen near this upper bound through the convenient implementation  $p_m^{(s)} = \left\lfloor \frac{L}{2} \cdot \frac{1.76 \gamma f_L M}{N_t B} \right\rfloor$ , the sweep parameter  $\theta'_p(s)$  is first determined to satisfy the hole-free drift requirement:

$$\theta'_p(s) = \frac{1.76 \gamma f_L M}{N_t B} - \frac{2}{L} p_m^{(s)}. \quad (62)$$

This choice fixes the sector's frequency-dependent sweep strength. The resulting scan is then placed within the desired angular interval of sector  $s$  by enforcing the target direction at the last subcarrier. Let  $\theta_M^{(s)}$  denote the desired direction associated with subcarrier  $m = M$  in sector  $s$ . Using (56) at  $m = M$  together with (61), we obtain

$$\theta'_t(s) = \frac{\theta_M^{(s)} - \frac{f_c}{f_M} S^{(s)}}{\rho_\theta}. \quad (63)$$

Finally, the desired starting direction at the first subcarrier is enforced by selecting the integer branch index  $p_1^{(s)}$ . Let  $\theta_1^{(s)}$  denote the desired direction at  $m = 1$ . Setting  $m = 1$  at (56) gives

$$\theta_1^{(s)} = \rho_\theta \theta'_t(s) + \frac{f_c}{f_L} \left( \theta'_p(s) + \frac{2}{L} p_1^{(s)} \right), \quad (64)$$

which can be rearranged to compute  $p_1^{(s)}$  as

$$p_1^{(s)} = \left\lfloor \frac{L}{2} \left( \frac{f_L}{f_c} (\theta_1^{(s)} - \rho_\theta \theta'_t(s)) - \theta'_p(s) \right) \right\rfloor, \quad (65)$$

where  $p_1^{(s)}$  denotes the nearest integer to (65).

## C. MINIMUM NUMBER OF PILOTS PER SECTOR UNDER SUB-ARRAY

After obtaining the sector-dependent angle parameters  $(\theta'_t(s), \theta'_p(s))$  in previous Subsection IV-B, the remaining task is to guarantee continuous coverage over the distance/curvature domain  $\alpha$  within each angular sector. Under the sub-array TTD-PS architecture, the coherent focusing points across subcarriers form periodic families indexed by  $(p_m, q_m)$ . In particular, the distance focusing relation on subcarrier  $m$  can be expressed as

$$\alpha_m = \rho_\alpha \alpha'_t + \frac{f_c}{f_m} \alpha'_p + \frac{\lambda_m}{L^2 d^2} q_m,$$

$$= \rho_\alpha \alpha'_t + \frac{f_c}{f_m} \left( \alpha'_p + \frac{2}{L^2 d^2} q_m \right). \quad (66)$$

To simplify the design along a fixed distance-alias branch, we fix  $q_m = q$  and define the composite distance-slope term

$$U \triangleq \alpha'_p + \frac{2}{L^2 d} q. \quad (67)$$

Hence, the frequency-driven sweep of the effective range/curvature focus across subcarriers is governed mainly by  $U$ , while  $\alpha'_t$  shifts the sweep vertically in the  $\alpha$ -domain.

$(\alpha'_p, q)$  is first chosen such that a single pilot sweep spans the target interval  $[\alpha_{\min}, \alpha_{\max}]$  over the band edges  $f_L < f_H$ . Since  $\alpha_m$  is largest at  $f_L$  and smallest at  $f_H$ , it is sufficient to enforce the band-edge coverage conditions

$$\rho_\alpha \alpha'_t + \frac{f_c}{f_H} U \leq \alpha_{\min}, \quad \text{and} \quad \rho_\alpha \alpha'_t + \frac{f_c}{f_L} U \geq \alpha_{\max}. \quad (68)$$

Subtracting the two inequalities yields the fundamental slope requirement  $\left(\frac{f_c}{f_L} - \frac{f_c}{f_H}\right) U \geq \alpha_{\max} - \alpha_{\min}$ . A convenient and tight design sets the inequality to equality as

$$U = \frac{\alpha_{\max} - \alpha_{\min}}{\frac{f_c}{f_L} - \frac{f_c}{f_H}}, \quad \text{and} \quad \alpha'_p = U - \frac{2}{L^2 d} q, \quad (69)$$

where  $q \in \mathbb{Z}$  is chosen so that  $\alpha'_p$  remains in a practical range. Once  $U$  is fixed, the feasible intercept range for  $\alpha'_t$  follows directly from (68) as

$$\alpha'_t \in \left[ \frac{1}{\rho_\alpha} \left( \alpha_{\max} - \frac{f_c}{f_L} U \right), \frac{1}{\rho_\alpha} \left( \alpha_{\min} - \frac{f_c}{f_H} U \right) \right]. \quad (70)$$

Although the above choice ensures that a single pilot sweeps across the desired distance interval, sub-array grouping may still create distance-domain gaps between adjacent strip trajectories. To quantify the worst-case gap, we introduce a sector-dependent composite distance term and derive an explicit bound in this subsection.

Accordingly, consider two subcarriers whose focused angles are adjacent under angular wrapping in  $\theta \in [-1, 1]$  (one near  $\theta_a \approx 1$  and the other near  $\theta_b \approx -1$ ), so that  $\Delta\theta \triangleq |\theta_b - \theta_a| \approx 2$ . Using the small-bandwidth approximation  $\left| \frac{1}{f_b} - \frac{1}{f_a} \right| \approx \frac{\Delta f}{f_c^2}$ , the corresponding focusing-point differences satisfy  $\Delta\theta \approx \frac{S_m^{(s)}}{f_c} \Delta f$  and  $\Delta\alpha \approx \frac{U}{f_c} \Delta f$ . Eliminating  $\Delta f$  yields the distance gap between two adjacent wrap-around strips as  $\Delta\alpha \approx \frac{U}{S_m^{(s)}} \Delta\theta \approx \frac{2U}{S_m^{(s)}}$ . Since this gap is maximized when the angular slope term is smallest,  $S_{\min}^{(s)} \triangleq \min S_m^{(s)}$  is defined, which leads to the conservative upper bound  $\Delta\alpha_{\text{upper}}^{(s)} \approx \frac{2U}{S_{\min}^{(s)}}$ . Then this worst-case gap is translated into the required number of interleaved distance pilots in sector  $s$ , letting  $\eta_f^\alpha$  denote the distance-domain 3-dB beamwidth at frequency  $f$  (derived in Appendix D), it is enforced that the uncovered gap does not exceed the overlap capability at the band edge via  $\Delta\alpha_{\text{upper}}^{(s)} \leq 2\eta_{f_H}^\alpha$ , where  $\eta_{f_H}^\alpha = \frac{4\beta_{3\text{dB}}^2 f_c^2}{N_t^2 c f}$  is obtained by substituting  $f = f_H$  into the closed-form 3 dB expression in Appendix D.

Therefore, the minimum number of distance pilots needed in sector  $s$  is

$$K_\alpha^{(s)} \geq \left\lceil \frac{\Delta\alpha_{\text{upper}}^{(s)}}{2\eta_{f_H}^\alpha} \right\rceil = \left\lceil \frac{U}{S_{\min}^{(s)} \eta_{f_H}^\alpha} \right\rceil. \quad (71)$$

**Algorithm 1** Grouped TTD-PS Parameter Manipulation for Sectorized Distance-Dependent Beam-Split Training

**Require:**  $N_t, f_c, B, M, d, L, [\theta_{\min}, \theta_{\max}], [\alpha_{\min}, \alpha_{\max}], \gamma$ .

- 1: Compute grouping-dependent coefficients and obtain  $\rho_\theta$  and  $\rho_\alpha$ .
- 2: Compute the strong-gain angular width  $W_{\text{strong}}^{(\text{pilot})}$  from (53).
- 3: Set  $\theta_{\text{tot}} \leftarrow \theta_{\max} - \theta_{\min}$  and obtain  $N_{\text{sec}}$  by (54).
- 4: **for**  $s = 1, \dots, N_{\text{sec}}$  **do**
- 5:   Sector shift using periodicity
- 6:    $\bar{\theta}_1^{(s)} \leftarrow \theta_{\min} + (s-1)W_{\text{strong}}^{(\text{pilot})}$
- 7:    $\bar{\theta}_M^{(s)} \leftarrow \theta_{\max} + (s-1)W_{\text{strong}}^{(\text{pilot})}$
- 8:   Map  $\bar{\theta}_1^{(s)}$  and  $\bar{\theta}_M^{(s)}$  into  $[-1, 1]$  and obtain wrap-around indices  $p_1^{(s)}$  and  $p_M^{(s)}$ .
- 9:   Denote the wrapped end-points as  $\theta_1^{(s)}$  and  $\theta_M^{(s)}$ .
- 10:   Compute  $S^{(s)}$  by (61).
- 11:   Obtain  $\theta_p^{(s)}$  by (62).
- 12:   Obtain  $\theta_t^{(s)}$  by (63).
- 13: **end for**
- 14: Set  $q$  and obtain  $\alpha'_p$  and  $\alpha'_t$  using (69) and (70). **return**  $\left\{ (\theta_t^{(s)}, \theta_p^{(s)}) \right\}_{s=1}^{N_{\text{sec}}}, \alpha'_t, \alpha'_p$ .

Finally, combining sector shifting with distance interleaving, the total training overhead required to cover the full angular range is

$$K_{\text{total}} = \sum_{s=1}^{N_{\text{sec}}} K_\alpha^{(s)}. \quad (72)$$

If the same distance-parameter design is reused across sectors so that  $K_\alpha^{(s)} = K_\alpha$ , then the overhead simplifies to  $K_{\text{total}} = N_{\text{sec}} K_\alpha$  pilots. The parameter-manipulation procedure for sector shifting and distance interleaving is summarized in Algorithm 1.

#### D. COMPLEXITY DISCUSSION

The complexity of the proposed grouped DDBS scheme should be interpreted from both training and hardware viewpoints. The training complexity is measured by the pilot overhead  $K_{\text{total}} = N_{\text{sec}} K_\alpha^{(s)}$ , where  $N_{\text{sec}}$  denotes the number of sector-shifted angle configurations and  $K_\alpha^{(s)}$  denotes the number of distance-interleaved pilots within each sector. Hence, the proposed method avoids exhaustive 2D polar sweeping while keeping the required training budget analytically controllable. Under the simulation setting used in Section VII,  $N_t = 256$ ,  $L = 2$ ,  $N_{\text{sec}} = 4$ , and  $K_\alpha^{(s)} = 3$ , which gives  $K_{\text{total}} = 12$  pilots. As summarized in Table 2, this overhead is substantially smaller than exhaustive and hierarchical near-field search, while remaining in the low-overhead regime of beam-split-based wideband training.



TABLE 1. Comparison of Beam Training Methods.

| Contents                                                 | This work  | [9], [15], [20] | [22]       | [23]       | [24]       | [16]–[18]    | [14]      |
|----------------------------------------------------------|------------|-----------------|------------|------------|------------|--------------|-----------|
| Near-field / Far-field training                          | Near-field | Near-field      | Near-field | Near-field | Near-field | Hybrid-field | Far-field |
| Wideband / frequency-dependent beamforming               | ✓          | ×               | ✓          | ✓          | ✓          | ×            | ✓         |
| Beam split leveraged for fast training                   | ✓          | ×               | ✓          | ✓          | ✓          | ×            | ✓         |
| Joint angle–range probing with pilot-efficient training  | ✓          | ✓               | ×          | ✓          | ✓          | ✓            | ×         |
| Fewer TTDs compared with the number of transmit antennas | ✓          | ×               | ×          | ×          | ×          | ×            | ✓         |
| Off-grid refinement (match-filter)                       | ✓          | ×               | ×          | ✓          | ×          | ×            | ×         |

TABLE 2. Quantitative Comparison of Training Overhead and Delay-Hardware Requirement for Representative Beam-Training Schemes.

| Scheme                                      | Training-overhead expression                       | Pilot overhead | Number of TTD units | Hardware implication                                                                                     |
|---------------------------------------------|----------------------------------------------------|----------------|---------------------|----------------------------------------------------------------------------------------------------------|
| Near-field exhaustive beam training* [15]   | $N_t S_{sw}$                                       | 2560           | not required        | No dedicated TTD hardware, but very large pilot burden                                                   |
| Near-field rainbow based beam training [22] | $S_{sw}$                                           | 10             | $N_t$               | Low pilot overhead, but high delay-hardware cost under per-antenna tuned-delay realization               |
| Far-field rainbow based beam training [14]  | 1                                                  | 1              | N/A                 | Very low training overhead, but it does not directly resolve near-field range                            |
| Near-field hierarchical beam training [18]  | $\sum_{\ell} U_{\ell} S_{\ell}$                    | 140            | not required        | No dedicated TTD hardware; multistage training cost remains substantial                                  |
| Inference from far-field beam training [19] | $N_t + S_{sw}$                                     | 266            | not required        | Lower overhead than exhaustive search, but pilot cost still scales with $N_t$                            |
| Near-field DDBS with per-antenna TTD [23]   | $K_{DDBS}$                                         | 3              | $N_t = 256$         | Very low pilot cost, but one tunable delay is required per antenna                                       |
| <b>Proposed grouped DDBS (AoSA TTD-PS)</b>  | $\mathbf{K}_{total} = N_{sec} \mathbf{K}_{\alpha}$ | <b>12</b>      | $N_t/L = 128$       | <b>Reduced delay-hardware cost and power consumption while maintaining practical pilot overhead [28]</b> |

The hardware complexity is governed mainly by the number of tunable TTD units. In a conventional per-antenna TTD-PS architecture, the delay count scales as  $N_{TTD} = N_t$ , whereas under the proposed AoSA implementation it becomes  $N_{TTD} = G = \frac{N_t}{L}$ . Therefore, delay sharing reduces the number of TTD units by a factor of  $L$ ; for the simulated case  $L = 2$ , the TTD count is reduced from 256 to 128. By contrast, the PS network still contains  $N_t$  phase shifters, so the hardware saving comes specifically from lowering the number of expensive tunable delays rather than from reducing the PS count. In addition to the number of TTD elements, sub-array sharing also lowers the associated control, routing, and calibration burden, although the exact implementation cost depends on the adopted RF platform and is therefore kept qualitative here.

Consequently, the proposed scheme does not minimize only pilots or only hardware. Instead, Table 2 shows that it operates between two extremes: PS-only schemes can avoid dedicated delay hardware but incur much larger training overhead, whereas per-antenna DDBS can use very few pilots but requires one TTD per antenna. The grouped DDBS design provides a balanced tradeoff between these two regimes.

Table 2 summarizes the training-overhead and delay-hardware requirements of representative beam-training schemes. In the table,  $S_{sw}$  denotes the number of sampled distance grids in the sweeping stage, and  $\sum_{\ell} U_{\ell} S_{\ell}$  denotes

the cumulative overhead of hierarchical search, where  $U_{\ell}$  and  $S_{\ell}$  are the angle and distance samples tested at layer  $\ell$ . For the representative setting  $N_t = 256$  and  $S_{sw} = 10$ , exhaustive beam training incurs the highest overhead of 2560 pilots, while near-field rainbow, hierarchical training, and inference from far-field training reduce it to 10, 140, and 266, respectively. Per-antenna DDBS further reduces the overhead to  $K_{DDBS} = 3$ , but at the expense of requiring one tunable TTD per antenna. In contrast, the proposed grouped DDBS requires  $K_{total} = N_{sec} K_{\alpha}^{(s)} = 12$  pilots, where  $N_{sec}$  denotes the number of sector shifts and  $K_{\alpha}^{(s)}$  denotes the number of distance-interleaved pilots in each sector, while reducing the number of TTD units to  $N_t/L = 128$  for  $L = 2$ . Hence, the proposed scheme achieves a more practical balance between pilot efficiency and hardware efficiency.

#### E. SELECTION OF THE SUB-ARRAY SIZE $L$

The sub-array size  $L$  is treated as a design parameter that governs the hardware–overhead–performance tradeoff of the proposed grouped TTD-PS architecture. Under the equal-size AoSA model considered in this work, the  $N_t$  antennas are partitioned into  $G = N_t/L$  contiguous sub-arrays, each equipped with one shared TTD. For a given  $L$ , the single-pilot strong-gain sector width is determined by (53), the required number of sector shifts is determined by (54), and the minimum number of distance pilots in each sector is

determined by (71). Hence, the total pilot overhead depends explicitly on  $L$  as

$$K_{\text{total}}(L) = \sum_{s=1}^{N_{\text{sec}}(L)} K_{\alpha}^{(s)}(L). \quad (73)$$

Meanwhile, the number of required TTD units is

$$N_{\text{TTD}}(L) = \frac{N_t}{L}. \quad (74)$$

Accordingly, the choice of  $L$  should jointly account for hardware reduction, required training overhead, and achievable beam-training performance. In this work, we adopt the following hardware-oriented selection criterion:

$$\begin{aligned} L^* &= \arg \min_L \frac{N_t}{L} \\ \text{s.t. } R(L) &\geq R_{\min}, \quad K_{\text{total}}(L) \leq K_{\max}, \end{aligned} \quad (75)$$

where  $R(L)$  denotes the achievable rate obtained by the proposed beam training for sub-array size  $L$ ,  $R_{\min}$  is the minimum required rate, and  $K_{\max}$  is the maximum affordable pilot budget. Therefore,  $L^*$  corresponds to the most hardware-efficient grouping factor that still satisfies the desired performance and training constraints.

The tradeoff induced by  $L$  follows directly from (53), (54), and (71). In particular, the one-pilot strong-gain sector width  $W_{\text{strong}}^{(\text{pilot})}(L)$  decreases approximately as  $1/L$ , which increases the required number of sector shifts  $N_{\text{sec}}(L)$ . Since the minimum distance-pilot requirement does not decrease sufficiently to offset this effect, the total pilot overhead  $K_{\text{total}}(L)$  also increases with  $L$ . By contrast, the number of required TTD units decreases as  $N_t/L$ . Therefore, increasing  $L$  reduces the delay-hardware cost, but it also narrows the strong-gain sector, increases the required training overhead, and aggravates the grouped-delay mismatch. Hence, the preferred operating point is determined by a hardware-overhead-performance tradeoff rather than by hardware reduction alone. The above criterion is kept general; its numerical instantiation for the representative simulation setup is provided in Section VII, where  $L = 2$  is shown to offer a practical operating point.

## VI. ROBUST REFINEMENT: MATCH-FILTER ENHANCEMENT

The sector-shifting and distance-interleaving design in Sec. V guarantees wideband coverage of the target  $(\theta, \alpha)$  search region under Sub-array grouping. In practice, however, finite SNR and grid mismatch may degrade the peak selection based on coarse codebook scanning. To improve robustness without additional pilot transmissions, one refinement mechanism is incorporated that is consistent with our training structure: a template-based match filter (MF) method. This method operates *after* obtaining a coarse estimate  $(\hat{\theta}, \hat{\alpha})$  from the proposed multi-strip training.

### A. GAIN-SIGNATURE MATCH FILTER (MF) REFINEMENT

**Offline stage (template construction).** Consider a fine candidate set around the coarse estimate  $(\hat{\theta}, \hat{\alpha})$ ,  $\mathcal{G}_{\text{fine}} \triangleq$

$\{(\theta_u, \alpha_v)\}_{u=1, v=1}^{N_\theta, N_\alpha}$ , where  $N_\theta$  and  $N_\alpha$  denote the numbers of sampled points in the angle and distance (curvature) domains, respectively. For each candidate  $(\theta_u, \alpha_v)$  and each designed training beamforming vector  $\{\mathbf{v}_m^{(k)}\}$ , the predicted *array-gain magnitude* can be computed as

$$a_{u,v}(m, k) \triangleq |\mathbf{c}_m^H(\theta_u, \alpha_v) \mathbf{v}_m^{(k)}|, \quad (76)$$

where  $\mathbf{c}_m(\theta, \alpha)$  is the polar-domain steering vector. Stacking the gains across all subcarriers and pilots yields the location-dependent signature

$$\mathbf{a}_{u,v} \triangleq [a_{u,v}(1, 1), \dots, a_{u,v}(M, 1), \dots, a_{u,v}(1, K_{\text{tot}}), \dots, a_{u,v}(M, K_{\text{tot}})]^T. \quad (77)$$

Since (76)–(77) depend only on the known training beams and the chosen grid, all templates can be precomputed and stored.

**Online stage (similarity matching).** Let  $\chi_m^{(k)}$  denote the received measurement on subcarrier  $m$  during the  $k$ -th training slot (pilot transmission), for  $m = 1, \dots, M$  and  $k = 1, \dots, K_{\text{tot}}$ . From the received pilots, the measured gain-signature vector is formed as

$$\hat{\mathbf{a}} \triangleq [|\chi_m^{(1)}|, \dots, |\chi_M^{(1)}|, \dots, |\chi_1^{(K_{\text{tot}})}|, \dots, |\chi_M^{(K_{\text{tot}})}|]^T. \quad (78)$$

A similarity score is then computed between the measured signature and each candidate template. A convenient choice is the normalized inner product (cosine similarity)  $\eta_{u,v} \triangleq \frac{\mathbf{a}_{u,v}^T \hat{\mathbf{a}}}{\|\mathbf{a}_{u,v}\|_2 \|\hat{\mathbf{a}}\|_2}$ , which mitigates sensitivity to unknown path-gain scaling. Finally, the MF-refined estimate is obtained by  $(\hat{\theta}_{\text{MF}}, \hat{\alpha}_{\text{MF}}) = \arg \max_{(\theta_u, \alpha_v) \in \mathcal{G}_{\text{fine}}} \eta_{u,v}$ . This refinement is particularly effective when the coarse solution is perturbed by noise or when the true user location is off-grid, because it leverages the entire wideband gain pattern (across  $m$  and  $k$ ) as a robust location fingerprint.

### B. COMPUTATIONAL AND STORAGE OVERHEAD OF THE MF REFINEMENT

The proposed MF refinement improves robustness by exploiting the entire wideband gain signature, but it also introduces additional local post-processing and memory compared with the coarse codebook-based estimator. Consider the local refinement window  $\mathcal{G}_{\text{fine}} = \{(\theta_u, \alpha_v)\}_{u=1, v=1}^{N_\theta, N_\alpha}$ . In the offline stage, one template vector  $\mathbf{a}_{u,v} \in \mathbb{R}^{MK_{\text{total}}}$  is constructed for each candidate location. If each template entry  $a_{u,v}(m, k) = |\mathbf{c}_m^H(\theta_u, \alpha_v) \mathbf{v}_m^{(k)}|$  is evaluated directly using (76), the one-time template-generation cost scales as  $\mathcal{O}(N_\theta N_\alpha MK_{\text{total}} N_t)$ . Since the MF stores magnitude-only signatures, the corresponding storage requirement scales as  $\mathcal{O}(N_\theta N_\alpha MK_{\text{total}})$  real numbers. Moreover, this offline cost depends only on the known training beams and the selected refinement grid, and can therefore be precomputed once for a given beam configuration.

In the online stage, after forming the measured signature  $\hat{\mathbf{a}} \in \mathbb{R}^{MK_{\text{total}}}$  from the received pilots, the cosine-similarity score for each candidate requires one inner product of length  $MK_{\text{total}}$ . Hence, if the template norms  $\|\mathbf{a}_{u,v}\|_2$  are pre-stored offline, the online search complexity scales as

$\mathcal{O}(N_\theta N_\alpha M K_{\text{total}})$ , followed by a simple maximum search over the  $N_\theta N_\alpha$  candidates. Importantly, this refinement does not increase the pilot overhead because it reuses the same  $K_{\text{total}}$  measurements already collected during beam training. Furthermore, since the search is restricted to a local neighborhood around the coarse estimate rather than the full polar-domain codebook, the practical online burden remains moderate. Therefore, the MF stage should be viewed as an optional robustness-enhancement step that trades additional local computation and storage for improved off-grid accuracy and better robustness at low SNR.

## VII. SIMULATION RESULTS

### A. SIMULATION SETUP

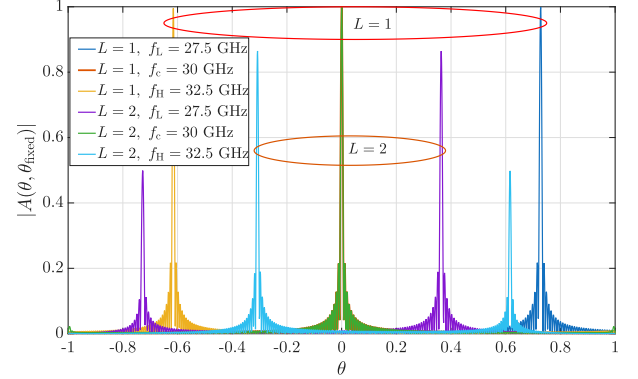
We evaluate the proposed sector-shifted distance-dependent beam-split (DDBS) training under the sub-array TTD-PS architecture and compare it with other beam training (BT) methods. Unless otherwise stated, all simulation results reported below are averaged over 500 Monte Carlo trials. Unless otherwise stated, the sector-wise training parameters are constructed following Algorithm 1.

Unless otherwise stated, we consider a downlink wideband near-field XL-MIMO OFDM system where the BS employs an  $N_t = 256$ -element ULA with carrier frequency  $f_c = 30$  GHz and bandwidth  $B = 5$  GHz over  $M = 1024$  subcarriers, with half-wavelength antenna spacing  $d = \lambda_c/2$ . This choice is motivated by the hardware-overhead-performance tradeoff analyzed in subsection V-E, where  $L = 2$  is shown to provide a representative operating point for the considered system parameters by halving the TTD count while maintaining near-ungrouped performance. The user azimuth angle  $\vartheta \in [-\pi/2, \pi/2]$  is mapped to the direction-cosine domain via  $\theta = \sin(\vartheta)$ , so that  $\theta \in [-1, 1]$ , and the distance interval  $[\mathcal{D}_{\min}, \mathcal{D}_{\max}] = [5 \text{ m}, 200 \text{ m}]$ , which maps to the distance-ring interval  $[\alpha_{\min}, \alpha_{\max}] = [0.0025, 0.1]$ . We adopt grouped delays with sub-array size  $L = 2$ , yielding  $G = N_t/L$  tunable delay units. Due to intra-sub-array coherence constraints, a fixed TTD-PS configuration concentrates the strong gain within a limited angular region. Therefore, the search range  $\theta \in [-1, 1]$  is partitioned into  $N_{\text{sec}}$  sectors, where  $N_{\text{sec}}$  is obtained by applying the analytical adjustable formula in (54); for  $L = 2$ , this yields  $N_{\text{sec}} = 4$ . Sector shifting is then applied by reconfiguring the angle-domain DDBS parameters across sector indices. Within each sector, *distance interleaving* is further employed to ensure continuous coverage in the distance/curvature domain, where the number of distance pilots per sector  $K_\alpha$  is obtained by applying the analytical adjustable formula in (71); for  $L = 2$ , this gives  $K_\alpha = 3$ . Consequently, where the number of pilots per sector is equally the total training overhead is  $K_{\text{total}} = N_{\text{sec}} K_\alpha = 12$  pilots. To avoid scanning holes between adjacent subcarriers, the oversampling parameter  $\gamma$  is set to 0.9.

**Angle-parameter construction:** For the first sector,  $(\theta_t^{(1)}, \theta_p^{(1)})$  is obtained by imposing boundary focusing at the band edges, i.e., setting the last subcarrier to  $\theta_M = 1$  and the first subcarrier to  $\theta_1 = -1$ . Substituting these constraints into

**TABLE 3.** Adjustable Sector-Wise TTD-PS Configuration  $\theta_t^{(s)}, \theta_p^{(s)}, \alpha'_t, \alpha'_p$  for Sub-Array DDBS Beam Training.

| Sector $s$ | $\theta_t^{(s)}$ | $\theta_p^{(s)}$ | $\gamma$ | $\alpha'_t$ | $\alpha'_p$ |
|------------|------------------|------------------|----------|-------------|-------------|
| 1          | -31.797          | 28.4             | 0.9      | -0.533      | 0.158       |
| 2          | -33.84           | 31.93            | 0.9      | -0.533      | 0.158       |
| 3          | -33.545          | 31.61            | 0.9      | -0.533      | 0.158       |
| 4          | -32.102          | 28.51            | 0.9      | -0.533      | 0.158       |



**FIGURE 3.** Normalized near-field array gain versus angle  $\theta$  at fixed  $\alpha = \alpha_m$ , evaluated at three subcarrier frequencies  $\{f_L, f_c, f_H\}$  (comparison: ungrouped  $L = 1$  vs. grouped/sub-array  $L = 2$ ).

(63) and (62) yields  $\theta_t^{(1)} = -31.797$  and  $\theta_p^{(1)} = 28.4$ . For the second sector, inverse scanning is applied by swapping the boundary assignments (first subcarrier at  $\theta_L = 1$  and last at  $\theta_M = -1$ ), resulting in  $\theta_t^{(2)} = -33.84$  and  $\theta_p^{(2)} = 31.93$ . The remaining sectors are generated by exploiting the angular periodicity of the DDBS trajectory and applying a  $45^\circ$  shift to cover the remaining angular intervals. The final sector-wise parameters used in simulations are summarized in Table 3.

**Distance parameters:** The distance-domain DDBS parameters are fixed across all sectors as  $\alpha'_t = -0.533$ ,  $\alpha'_p = 0.158$ , so that only the angular placement of the strong-gain region changes from sector to sector, while the distance-sweep law remains identical.

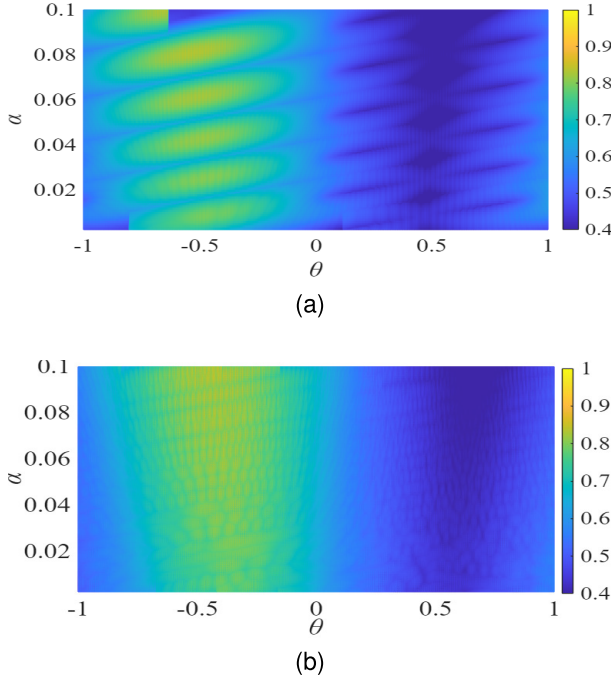
### B. BEAM TRAINING AND PERFORMANCE EVALUATION

As a performance metric, the average achievable rate (bit/s/Hz) is evaluated using the beamforming vector returned by beam training:

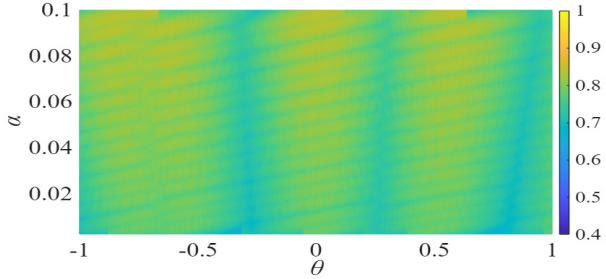
$$R = \frac{1}{M} \sum_{m=1}^M \log_2 \left( 1 + \frac{P_t}{\sigma^2} |\mathbf{g}_m^T \mathbf{v}_m|^2 \right), \quad (79)$$

To begin, we validate the proposed sub-array TTD-PS beamforming vector, Fig. 3 plots normalized 1D array-gain slices versus  $\theta$  for three representative subcarriers  $\{f_L, f_c, f_H\}$ , where the curvature is fixed at  $\alpha_{\text{fixed}} = \alpha_m$  given by the LS focusing mapping.

Compared with the ungrouped architecture ( $L = 1$ ), the grouped/sub-array design ( $L = 2$ ) shares one TTD within each sub-array, which introduces a frequency-dependent residual intra-sub-array phase mismatch. As a result, the dominant peak exhibits a reduced height and a frequency-dependent shift. Moreover, additional replicated peaks appear due to aliasing induced by the inter-sub-array spacing  $Ld$ ;



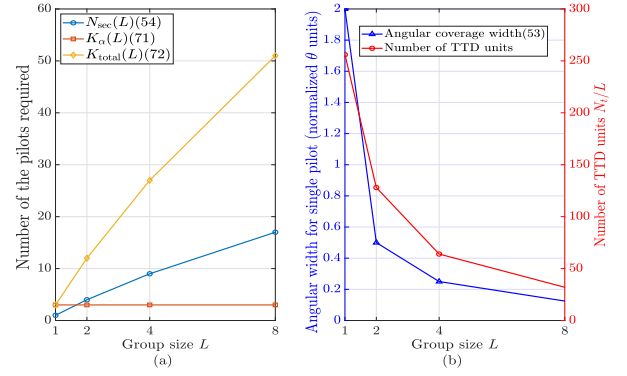
**FIGURE 4.** DDBS gain maps under Sub-array TTD-PS ( $L = 2$ ). (a) One pilot yields strong gain only in a limited angular sector. (b) Three distance-interleaved pilots improve  $\alpha$  (range/curvature) coverage while keeping the same angular sector.



**FIGURE 5.** Overall gain map with four-sector coverage under Sub-array TTD-PS ( $L = 2$ ), where sector shifting spans the target  $(\theta, \alpha)$  region.

their locations vary with frequency since the alias spacing scales with  $\lambda_m$ . The replica levels also change across subcarriers because the intra-sub-array coherence acts as a frequency-dependent mask that attenuates different alias branches. After applying the proposed LS-based focusing design, the dominant peak gain increases to around 0.9 of the ideal maximum, while the remaining weaker periodic peaks are still observed, consistent with the aliasing effect of spacing  $Ld$ .

Fig. 4 illustrates the distance-dependent beam-splitting behavior under the sub-array TTD-PS architecture ( $L = 2$ ). With a fixed angle-parameter configuration  $(\theta'_t, \theta'_p)$ , the beamforming vector produces a concentrated strong-gain region confined to a limited angular sector. As shown in Fig. 4(a), using a single pilot yields strong gain only over a narrow sector and a limited curvature (range) span. When the number of distance-interleaved pilots is increased to three, Fig. 4(b) shows that the strong-gain region expands along the  $\alpha$  dimension (improved range/curvature coverage); however, the angular support remains confined to essentially the same



**FIGURE 6.** Pilot overhead, angular coverage width, and hardware cost versus group size  $L$ . Subplot (a) shows the required pilot counts, while subplot (b) shows the one-pilot angular coverage width and the number of TTD units. In (b), a width of 2 indicates full angular coverage over  $\theta \in [-1, 1]$ .

sector. Therefore, increasing distance-interleaved pilots alone cannot provide uniform gain over the full angular range, which motivates sector shifting by reconfiguring  $\theta'_t(s), \theta'_p(s)$  across multiple sector indices, as demonstrated in Fig. 5.

The required number of sector shifts is determined by the sector-count bound in (54). For  $L = 2$  and  $\theta \in [-1, 1]$  under the 3-dB strong-gain criterion, the bound suggests about four sectors; accordingly, we set  $N_{\text{sec}} = 4$  to provide margin against discretization, subcarrier variations, and edge effects. Within each sector, multiple pilots are still needed to cover the curvature dimension, and we allocate three distance-interleaved pilots per sector. Hence, the total training overhead is  $K_{\text{total}} = N_{\text{sec}} \times 3 = 12$  pilots. The resulting overall gain map in Fig. 5 confirms that the proposed four-sector design provides strong coverage across the target  $(\theta, \alpha)$  region under the sub-array TTD-PS architecture ( $L = 2$ ).

Fig. 6 illustrates the analytical dependence of the grouped-training design on the sub-array size  $L$ . In Fig. 6(a), increasing  $L$  increases the required number of sector shifts  $N_{\text{sec}}(L)$  and, consequently, the total pilot overhead  $K_{\text{total}}(L)$ , while the minimum distance-pilot requirement per sector  $K_{\alpha}^{(s)}(L)$  changes only mildly. This is because a larger  $L$  reduces the one-pilot angular coverage width and therefore requires more sector shifts to maintain full angular coverage. In Fig. 6(b), the one-pilot angular width decreases with  $L$ , whereas the number of required TTD units decreases as  $N_t/L$ , reflecting the hardware saving achieved by delay sharing. Note that the angular width is measured in the normalized spatial-angle variable  $\theta = \sin(\vartheta)$ , where  $\vartheta \in [-90^\circ, 90^\circ]$  and thus  $\theta \in [-1, 1]$ ; hence, a value of 2 corresponds to full angular coverage. In particular, for  $L = 1$ , the masking effect disappears and one pilot is regarded as covering the full angular search range. Therefore, the grouped architecture exhibits a clear hardware-training tradeoff: larger  $L$  is more hardware-efficient, but requires a larger training budget to guarantee complete angle-distance coverage. To evaluate how this analytical tradeoff translates into end performance, the achievable rate is next examined as a function of  $L$ .

Fig. 7 shows the achievable rate versus the group size  $L$  when each  $L$  is assigned its own analytically determined



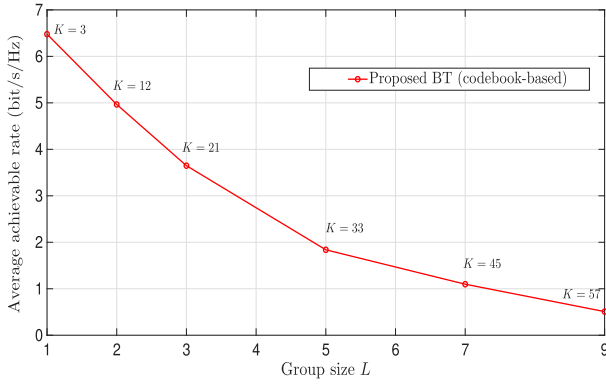


FIGURE 7. Achievable rate versus group size  $L$  using the analytically determined pilot budget  $k(L) = K_{\text{total}}(L)$ .

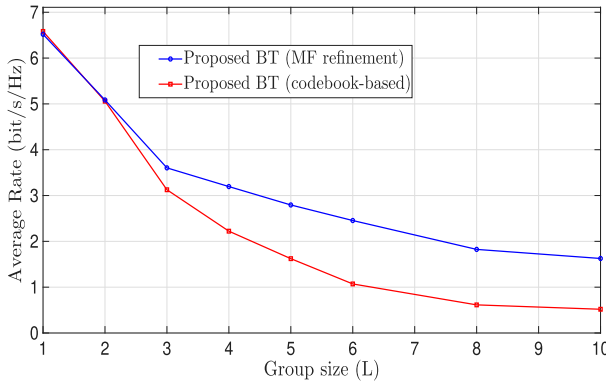


FIGURE 8. Achievable-rate performance versus sub-array size  $L$  where  $K = 12$  pilots.

pilot budget  $k(L) = K_{\text{total}}(L)$ . Although larger  $L$  is allocated more pilots to maintain full coverage, the average rate still decreases markedly with increasing  $L$ . Since the average rate is evaluated using the beamforming vector returned by beam training, this trend indicates that the end-to-end beam-training performance deteriorates at large  $L$  mainly because grouped TTD sharing introduces stronger intra-sub-array phase mismatch and weaker coherent beam focusing. In other words, the observed loss is not primarily caused by under-allocating pilots, but by the hardware-induced degradation that limits the final beam selected after training.

Fig. 8 studies the effect of the grouping factor (sub-array size)  $L$  on the achievable rate under the proposed grouped TTD-PS architecture. As  $L$  increases, fewer TTD elements are available ( $G = N_t/L$ ), which reduces delay-control resolution and limits accurate compensation of frequency-dependent near-field phase across the array.

As a result, intra-sub-array residual delay/phase mismatch increases, coherent combining degrades, and the beam energy spreads into unintended angle-distance regions. Therefore, the average rate decreases and saturates at a lower level for large  $L$ . In contrast,  $L = 2$  provides a favorable tradeoff, achieving a rate close to the ungrouped case while reducing the number of TTD elements by half.

Besides spectral efficiency, Fig. 8 also provides an explicit hardware-efficiency interpretation when read together with Table 2. In the proposed AoSA architecture, hardware

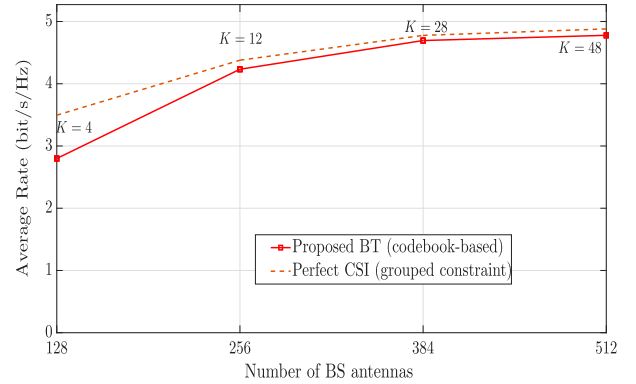
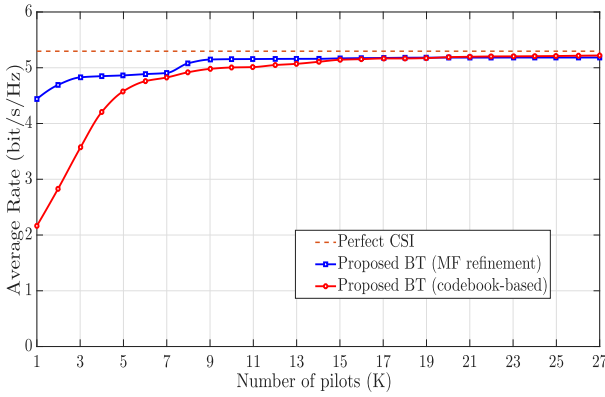


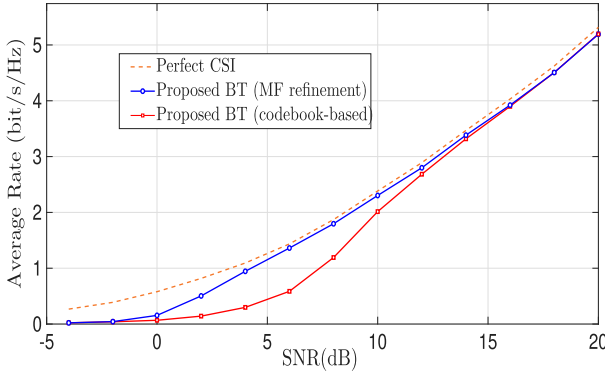
FIGURE 9. Achievable rate versus Number of BS antennas at SNR = 20 dB (sub-array size  $L = 2$ ).

efficiency is reflected by the reduced number of tunable TTD units, which scales as  $N_{\text{TTD}} = N_t/L$ , whereas spectral efficiency is reflected by the achievable-rate curves. Hence, increasing  $L$  decreases the delay-hardware cost but simultaneously increases the intra-sub-array mismatch. For the operating point used throughout this section, i.e.,  $N_t = 256$ ,  $L = 2$ ,  $N_{\text{sec}} = 4$ , and  $K_\alpha = 3$ , the proposed method requires  $K_{\text{total}} = N_{\text{sec}}K_\alpha = 12$  pilots and  $N_{\text{TTD}} = N_t/L = 128$  tunable delays, corresponding to a 50% reduction in TTD count relative to the ungrouped/per-antenna design. The fact that the corresponding achievable rate remains close to the ungrouped curve indicates that the proposed grouped DDBS architecture attains a practically useful tradeoff between hardware efficiency and spectral efficiency. Having identified this practical grouping configuration, we next examine how the proposed grouped beam training scales with the array size. Fig. 9 illustrates how the achievable rate of the proposed grouped beam training varies with the number of BS antennas, thereby demonstrating its scalability in XL-MIMO settings. The grouped architecture with fixed sub-array size  $L = 2$  is considered, so the number of TTD units scales as  $N_t/L$  as the array grows. For each array size, the pilot budget is selected according to the corresponding grouped-training design. It can be observed that the achievable rate of the proposed codebook-based method increases with  $N_t$  and remains close to the grouped perfect-CSI upper bound across the tested array sizes.

Fig. 10 depicts the average rate versus the pilot number  $K$  at SNR = 20 dB for the grouped architecture with  $L = 2$ . Under a limited pilot budget, the on-grid grouped method suffers a noticeable rate loss since a small  $K$  provides fewer interleaved strip shifts, leaving larger uncovered regions in the distance (curvature) domain. The proposed match-filter refinement improves performance when the pilot budget is limited by correlating the received multi-subcarrier measurements with a bank of precomputed focusing signatures, which mitigates off-grid mismatch and reduces erroneous strip selections. As  $K$  increases, both methods improve and approach the grouped perfect-CSI upper bound, so the performance gap becomes small once sufficient pilots are available. For this configuration, the performance is



**FIGURE 10.** Achievable rate versus pilot budget  $K$  at  $\text{SNR} = 20$  dB (sub-array size  $L = 2$ ).



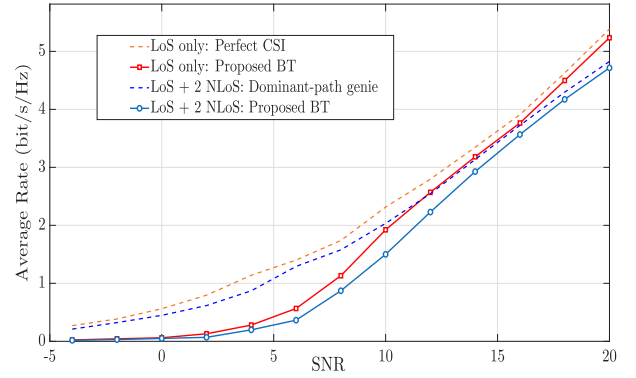
**FIGURE 11.** Achievable rate versus SNR with fixed pilot budget  $K = 12$ .

effectively saturated around  $K \approx 12$  pilots, which is adopted in the subsequent SNR experiments.

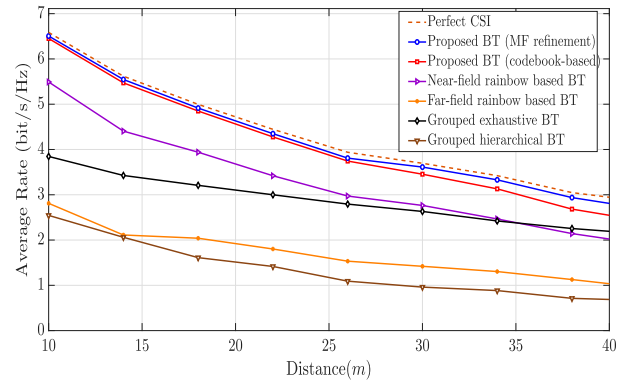
Fixing  $K = 12$  pilots, Fig. 11 shows the average rate versus SNR. At low SNR, all methods achieve near-zero rate because received pilots are noise-dominated and reliable strip/sector selection is difficult.

As SNR increases, the rate improves rapidly for all schemes, and the match-filter refinement consistently outperforms the on-grid method, especially in the medium-SNR region where both grid mismatch and noise affect selection. At high SNR, the proposed grouped solutions approach the grouped perfect-CSI curve, indicating that the dominant limitation becomes the sub-array hardware constraint rather than the training algorithm. To further evaluate robustness beyond the dominant-LoS benchmark, we next consider a LoS-dominant sparse multipath channel generated according to the general model in (2).

To further evaluate robustness beyond the dominant-LoS baseline, we consider a LoS-dominant multipath channel generated from the general model, consisting of one dominant LoS path and two reflected NLoS paths. Their relative powers are 0, -7, and -10 dB, respectively. Compared with the dominant LoS path, the two reflected paths have angular offsets of approximately  $+10^\circ$  and  $-12^\circ$ , distance offsets of +5 m and -4 m, and independent random phases. For fair comparison, each multipath realization is normalized to have the same Frobenius norm as the corresponding LoS-only realization.



**FIGURE 12.** Average achievable rate versus SNR for LoS-only and LoS-dominant sparse multipath channels under the proposed grouped DDBS beam training. Corresponding Perfect-CSI bounds are also shown.



**FIGURE 13.** Achievable rate versus user distance (10–40 m) accounting for path loss.

As shown in Fig. 12, the proposed grouped beam training in the multipath case follows the same overall SNR trend as in the LoS-only case and remains close to its corresponding dominant-path genie benchmark, while exhibiting a moderate and expected rate loss due to the NLoS reflected paths acting mainly as structured interference. Overall, the result confirms the robustness of the proposed AoSA grouped TTD-PS beam-training scheme in sparse near-field multipath channels.

After validating the proposed scheme under sparse near-field multipath propagation, we next investigate the effect of distance-dependent path loss. Accordingly, Fig. 13 evaluates the average achievable rate versus user distance, where the transmit-SNR ratio is fixed at 80 dB. To make the transmit-power setting explicit, the noise variance is computed from the thermal-noise power over the system bandwidth as  $\sigma^2 = k_B T_0 B F$ , where  $k_B = 1.380649 \times 10^{-23}$  J/K is Boltzmann's constant,  $T_0 = 290$  K is the reference temperature,  $B$  is the system bandwidth, and  $F$  is the receiver noise factor [30]. Assuming an ideal receiver noise figure, i.e.,  $F = 1$ , and using  $B = 5$  GHz, the noise power becomes  $\sigma^2 \approx 2.0 \times 10^{-11}$  W  $\approx -77$  dBm. Since the transmit-SNR ratio is defined as  $\text{SNR} = \frac{P_t}{\sigma^2}$ , the transmit power corresponding to transmit SNR = 80 dB is  $P_t = 2.0 \times 10^{-3}$  W = 2 mW  $\approx 3$  dBm. Fig. 13 evaluates the average achievable rate as the user distance increases from 10 m to 40 m when distance-dependent path loss is taken into account. As expected, all

achievable-rate curves decrease with distance because the received signal power becomes weaker as the propagation distance increases. Nevertheless, the proposed grouped near-field methods remain consistently superior to the practical baselines across the whole tested range. The MF refinement further improves performance by reducing the residual gap caused by on-grid selection and grouped-delay mismatch, thereby maintaining performance close to the perfect-CSI upper bound. By contrast, the far-field rainbow baseline performs noticeably worse because its far-field beam model is inherently mismatched to the near-field regime. The near-field rainbow baseline also remains below the proposed grouped DDBS schemes over the whole distance range.

To isolate the gain of the proposed training algorithm from that of the grouped hardware architecture, Fig. 13 also includes two grouped-hardware conventional baselines, namely grouped exhaustive BT and grouped hierarchical BT. Both baselines use the same grouped AoSA TTD-PS architecture with  $L = 2$  as the proposed method, so the comparison is carried out under the same grouped-hardware constraint and the observed differences mainly reflect the beam-training strategy rather than the hardware structure. It is seen that both grouped conventional baselines remain below the proposed grouped DDBS schemes across the whole considered distance range. This confirms that the performance gain comes not only from adopting grouped hardware, but also from the proposed sector-shifted and distance-interleaved grouped DDBS training design, which is better suited for efficient near-field wideband angle-distance probing under grouped-delay constraints.

### VIII. CONCLUSION

This paper developed a hardware-efficient extension of distance-dependent beam-split (DDBS) training for near-field wideband XL-MIMO by introducing an AoSA/sub-array TTD-PS architecture with grouped TTD elements. By analyzing the resulting angle-range periodicity, we showed that, under sub-array-shared TTDs, each pilot maintains strong gain only over a limited angular sector, which makes naively increasing the number of pilots insufficient to achieve full angular coverage. To address this, we derived 3-dB coverage-based design rules that (i) determine the one-pilot strong-gain sector width and the required number of sector shifts, and (ii) quantify distance-domain gaps and translate them into a minimum number of interleaved distance pilots, leading to a simple total-overhead expression.

Building on these rules, we proposed a sector-shifted, distance-interleaved grouped DDBS training algorithm, and further incorporated practical refinements: an LS-based focusing step that boosts the main-lobe gain to about 0.9 of the ideal maximum, and an off-grid match-filter refinement that improves robustness under limited pilot budgets and low-SNR conditions. Simulation results highlight the key architectural tradeoff: increasing the sub-array size  $L$  reduces the number of tunable delays but degrades coherent combining, whereas  $L = 2$  achieves a strong balance—cutting

the number of TTD elements by half while remaining close to the ungrouped rate, and reaching near-saturation around  $K \approx 12$  pilots. At high SNR, the residual gap to the ideal curve is dominated by the grouped hardware constraint rather than the training procedure. Finally, this work has focused on the single-user initial beam-training stage in order to isolate the impact of grouped TTD hardware on near-field wideband angle-distance probing. Extending the proposed grouped DDBS framework to multi-user XL-MIMO systems is an important direction for future work. Such an extension would require additional design components, including pilot coordination, inter-user interference management, user scheduling, and joint beam/resource allocation.

### APPENDIX A CLOSED-FORM SYMMETRY SUMS FOR CONTIGUOUS SYMMETRIC SUB-ARRAYS

Define  $S_2 \triangleq \sum_{\hat{n}=-N}^N \hat{n}^2$ ,  $S_4 \triangleq \sum_{\hat{n}=-N}^N \hat{n}^4$ ,  $C_1 \triangleq \sum_{\hat{n}=-N}^N \hat{n} \hat{n}_s$ , and  $C_2 \triangleq \sum_{\hat{n}=-N}^N \hat{n}^2 \hat{n}_s^2$ . The corresponding closed forms are  $S_2 = \frac{N(N+1)(2N+1)}{3}$  and  $S_4 = \frac{N(N+1)(2N+1)(3N^2+3N-1)}{15}$ , while  $C_1 = L^2 \sum_{\hat{n}_s=-Q}^Q \hat{n}_s^2$  and  $C_2 = L^3 \sum_{\hat{n}_s=-Q}^Q \hat{n}_s^4 + \frac{L(L^2-1)}{12} \sum_{\hat{n}_s=-Q}^Q \hat{n}_s^2$ . Moreover,  $\sum_{\hat{n}_s=-Q}^Q \hat{n}_s^2 = \frac{Q(Q+1)(2Q+1)}{3}$  and  $\sum_{\hat{n}_s=-Q}^Q \hat{n}_s^4 = \frac{Q(Q+1)(2Q+1)(3Q^2+3Q-1)}{15}$ .

### APPENDIX B PROOF OF PROPOSITION 1

*Proof:* In this appendix, we show that the far-field gain (obtained from the general gain expression (25) by setting  $\alpha = 0$ ) factorizes into the product form in (32) with the phase increments in (33). Step 1: Far-field steering vector and TTD-PS beamforming vector For  $\alpha = 0$ , the (normalized) far-field steering vector at subcarrier  $m$  is  $[\mathbf{u}_m(\theta, 0)]_{\hat{n}} = \frac{1}{\sqrt{N_t}} \exp(-jk_m \hat{n} d \theta)$ ,  $\hat{n} \in \{-N, \dots, N\}$ , where  $k_m = 2\pi f_m / c$  and  $N = (N_t - 1)/2$ . Under the sub-array TTD-PS model (TTD at  $k_m$  and PS at  $k_c$ ),  $[\mathbf{v}_m]_{\hat{n}} = \frac{1}{\sqrt{N_t}} \exp(-jk_m g(\hat{n}) L d \theta'_t) \exp(-jk_c \hat{n} d \theta'_p)$ , where  $g(\hat{n}) \triangleq [\hat{n}/L]$ , and  $\hat{n} = \hat{n}_s L + i$ . Step 2: Convert the gain to a double sum By definition,  $G_m(\theta, 0) \triangleq |\mathbf{u}_m^H(\theta, 0) \mathbf{v}_m|$ . Substituting (VIII) gives

$$G_m(\theta, 0) = \frac{1}{N_t} \left| \sum_{\hat{n}_s=-Q}^Q \sum_{i \in \mathcal{I}} \exp(jk_m(\hat{n}_s L + i) d \theta) \times \exp(-jk_m \hat{n}_s L d \theta'_t) \times \exp(-jk_c(\hat{n}_s L + i) d \theta'_p) \right|. \quad (80)$$

Using  $\hat{n} = \hat{n}_s L + i$  with  $g(\hat{n}) = \hat{n}_s$  yields

$$G_m(\theta, 0) = \frac{1}{N_t} \left| \sum_{\hat{n}_s=-Q}^Q \sum_{i \in \mathcal{I}} \exp(jk_m(\hat{n}_s L + i) d \theta) \times \exp(-jk_m \hat{n}_s L d \theta'_t) \times \exp(-jk_c(\hat{n}_s L + i) d \theta'_p) \right|. \quad (81)$$

Collecting the terms in  $\hat{n}_s$  and  $i$ , we obtain

$$\exp(jk_m(\hat{n}_s L + i) d \theta) \exp(-jk_m \hat{n}_s L d \theta'_t) \times \exp(-jk_c(\hat{n}_s L + i) d \theta'_p) \quad (82)$$

$$= \exp(j\hat{n}_s L d (k_m(\theta - \theta'_t) - k_c \theta'_p)) \times \exp(ji d (k_m \theta - k_c \theta'_p)). \quad (83)$$

Define (same as (33))

$$\Omega_m(\theta) \triangleq Ld(k_m(\theta - \theta'_t) - k_c\theta'_p), \quad (84)$$

$$z_m(\theta) \triangleq d(k_m\theta - k_c\theta'_p). \quad (85)$$

Then  $G_m(\theta, 0) = \frac{1}{N_t} \left| \sum_{\hat{n}_s=-Q}^Q \sum_{i \in \mathcal{I}} e^{j\hat{n}_s\Omega_m(\theta)} e^{ji z_m(\theta)} \right|$ . Since the inner sum does not depend on  $\hat{n}_s$ , the double sum factorizes as

$$\begin{aligned} \sum_{\hat{n}_s=-Q}^Q \sum_{i \in \mathcal{I}} e^{j\hat{n}_s\Omega_m(\theta)} e^{ji z_m(\theta)} &= \left( \sum_{\hat{n}_s=-Q}^Q e^{j\hat{n}_s\Omega_m(\theta)} \right) \\ &\quad \times \left( \sum_{i \in \mathcal{I}} e^{ji z_m(\theta)} \right). \end{aligned} \quad (86)$$

Therefore,

$$G_m(\theta, 0) = \frac{1}{N_t} \left| \sum_{\hat{n}_s=-Q}^Q e^{j\hat{n}_s\Omega_m(\theta)} \right| \left| \sum_{i \in \mathcal{I}} e^{ji z_m(\theta)} \right|, \quad (87)$$

which is exactly (32) with (84)–(85) (i.e., (33)). ■

## APPENDIX C PROOF OF PROPOSITION 2

*Proof:* From (25), after substituting  $\hat{n} = \hat{n}_s L + i$  and collecting the terms that depend only on the sub-array index  $\hat{n}_s$ , the outer summation can be written as  $\sum_{\hat{n}_s=-Q}^Q e^{j\phi_g(\hat{n}_s)}$  with  $\phi_g(\hat{n}_s) = \hat{n}_s L d x'_m - \hat{n}_s^2 L^2 d^2 y'_m$ , which matches (38). A sufficient condition for all phasors to align (hence maximizing the outer sum) is the modulo- $2\pi$  alignment of the linear and quadratic lattice terms:  $Ld x'_m = 2\pi p_m$ ,  $L^2 d^2 y'_m = 2\pi q_m$ , for some integers  $p_m, q_m \in \mathbb{Z}$  (as stated right before Proposition 1).

At a coherent focusing point  $(\theta_m, \alpha_m)$ , using the definitions in (25),  $x_m = k_m\theta_m - k_c\theta'_p$ ,  $y_m = k_m\alpha_m - k_c\alpha'_p$ , and using the LS-induced effective focusing reference in (29)–(31), the sub-array-level residuals relative to the LS effective focus can be written as  $x'_m = k_m\theta_m - k_c\theta'_p - k_m\rho\theta'_t$ ,  $y'_m = k_m\alpha_m - k_c\alpha'_p - k_m\rho\alpha'_t$ . Substituting these into the alignment conditions yields  $k_m\theta_m - k_c\theta'_p - k_m\rho\theta'_t = \frac{2\pi}{Ld} p_m$ ,  $k_m\alpha_m - k_c\alpha'_p - k_m\rho\alpha'_t = \frac{2\pi}{L^2 d^2} q_m$ .

Therefore,

$$\theta_m = \rho\theta'_t + \frac{k_c}{k_m}\theta'_p + \frac{2\pi}{k_m L d} p_m, \quad (88)$$

$$\alpha_m = \rho\alpha'_t + \frac{k_c}{k_m}\alpha'_p + \frac{2\pi}{k_m L^2 d^2} q_m. \quad (89)$$

Finally, using  $\frac{k_c}{k_m} = \frac{f_c}{f_m}$  and  $\frac{2\pi}{k_m} = \lambda_m$  gives (39)–(40). ■

## APPENDIX D DERIVATION OF THE CURVATURE-DOMAIN 3-DB BEAMWIDTH $\eta_\alpha$ UNDER AOSA TTD-PS

Consider subcarrier frequency  $f$  with wavenumber  $k = 2\pi f/c$  and wavelength  $\lambda = c/f$ . Along the curvature cut  $\theta = \bar{\theta}$ , let  $\alpha = \bar{\alpha} + \Delta\alpha$ . The (normalized) array gain then reduces to a quadratic-phase sum across the aperture,

$$G(0, \Delta\alpha) = \frac{1}{N_t} \left| \sum_{\hat{n}=-N}^N \exp(-j\hat{n}^2 d^2 k \Delta\alpha) \right|, \quad (90)$$

where  $N = (N_t - 1)/2$  and  $d = \lambda_c/2$  is the antenna spacing.

For large  $N_t$ , (90) is well-approximated by a Fresnel envelope  $G(\Delta\alpha) \approx F(\beta) \triangleq \frac{C(\beta) + jS(\beta)}{\beta}$ , where  $C(\beta) = \int_0^\beta \cos(\frac{\pi}{2} t^2) dt$  and  $S(\beta) = \int_0^\beta \sin(\frac{\pi}{2} t^2) dt$  are Fresnel integrals and  $\beta \triangleq \frac{N_t d}{\sqrt{\lambda}} \sqrt{|\Delta\alpha|}$  [23]. Defining the 3-dB point by  $F(\beta_{3dB}) = 1/\sqrt{2}$  gives  $\beta_{3dB} \approx 1.318$ . Hence, the one-sided curvature 3-dB beamwidth at frequency  $f$  is

$$\eta_\alpha^f \triangleq |\Delta\alpha|_{3dB} = \frac{\beta_{3dB}^2 \lambda}{N_t^2 d^2} = \frac{4\beta_{3dB}^2 f_c^2}{N_t^2 c f}, \quad (91)$$

where  $d = c/(2f_c)$  is used in the last equality.

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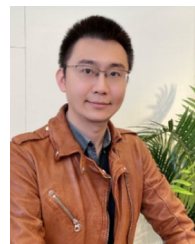
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